Real Time Simulation of Vehicle Dynamics

B.E. Project Report-B

Submitted in partial fulfillment of the requirements

For the degree of

Bachelor of Engineering

(Electronics Engineering)

by

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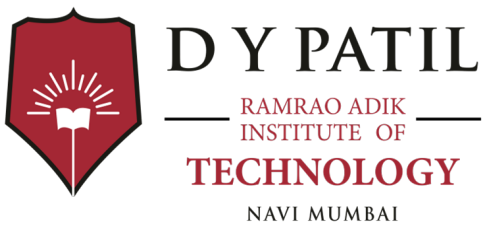
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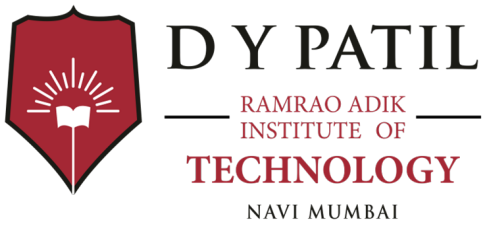
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April 2023



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Certificate

This is to certify that, the project report-B titled

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is a bonafide work done by

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Heramb Bhor (19EE1048)
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and is submitted in the partial fulfillment of the requirement for the degree of

Bachelor of Engineering

(Electronics Engineering)

to the

University of Mumbai.



Examiner 1	Examiner 2	Supervisor

Project Coordinator	Head of Department	Principal

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Abstract

In this work, our main objective is to simulate vehicle dynamics and control systems in real-time. Our team has conducted an in-depth study of different control systems including suspension, anti-lock braking (ABS), motor speed, throttle, and cruise control. We have also analyzed open and closed-loop systems to gain a better understanding of the control system's behavior. To implement the control systems, we have utilized Microlabbox and DSPACE, which are widely used by the automotive industry for prototyping and testing. Through simulation, we have observed that the control systems are highly effective in enhancing the vehicle's performance and safety. In conclusion, this project provides valuable insights into the design and implementation of advanced control systems for modern vehicles.

Keywords:
Transfer function, Vehicle Dynamics, Microlabbox, Real-time simulation, Control systems, Suspension system, ABS system, Motor speed, Throttle system, Cruise control, Microlabbox, DSPACE, Performance, Safety, Automotive industry.

Contents

Abstract **iii**

List of Figures **vi**

1 Introduction **1**
1.1 Motivation 1
1.2 Objective 1
1.3 Problem Definition 1
1.4 Organization of Report 2

2 Literature Survey **3**

3 System Design **6**
3.1 Overview of vehicle dynamics 6
3.2 Hardware Used 7
3.2.1 Microlabbox 7
3.2.2 Digital Storage Oscilloscope 8
3.3 Software Used 9
3.3.1 Matlab 9

4 System Implementation **10**
4.1 Suspension System 10
4.2 Cruise Control System 14
4.3 Antilock Braking System 18
4.4 Motor Speed System 22
4.5 Throttle System 26

5 Result and Analysis **30**
5.1 Suspension System 30
5.2 Cruise Control System 33
5.3 Antilock Braking System 36
5.4 Motor Speed System 38
5.5 Throttle System 40

6 Conclusion and Future Implementation **42**
6.1 Conclusion 42
6.2 Future Implementation 43

Bibliography **44**

Acknowledgments **46**

List of Figures

3.1 Dspace Microlabbox 7
3.2 Digital Storage Oscilloscope 8

4.1 Open Loop Suspension System 11
4.2 Closed Loop Suspension System 11
4.3 Dspace model of Suspension System 12
4.4 PID parameter of Suspension System 12
4.5 State Space Model 13
4.6 Suspension system - Statespace model block parameter 13
4.7 Open Loop Cruise Control System 14
4.8 Closed Loop Cruise Control System 15
4.9 Dspace model of Cruise Control System 15
4.10 PID parameter of Cruise control System 16
4.11 State Space Model 17
4.12 Cruise control system -Statespace model block parameter 17
4.13 Open Loop Antilock Braking System 18
4.14 Closed Loop Antilock Braking System 19
4.15 Dspace model of ABS System 19
4.16 PID parameter of anti lock breaking System 20
4.17 State Space Model 21
4.18 ABS - Statespace model block parameter 21
4.19 Open Loop Motor Speed System 22
4.20 Closed Loop Motor Speed System 23
4.21 Dspace model of motor speed System 23
4.22 PID parameter of motor speed system 24
4.23 State Space Model 25
4.24 Motor speed - Statespace model block parameter 25
4.25 Open Loop Throttle System 26
4.26 Closed Loop Throttle System 27
4.27 Dspace model of Throttle System 27
4.28 PID parameter of throttle system 28
4.29 State Space Model 29
4.30 Throttle system - Statespace model block parameter 29

5.1 Open loop suspension system response 31
5.2 Closed loop suspension system response 31
5.3 Closed loop suspension system response on dspace 32
5.4 Statespace response of suspension system on dspace 32

5.5 Open loop cruise control system response 33
5.6 Closed loop cruise control system response 33
5.7 Closed loop cruise control system response on dspace 34
5.8 Statespace response of cruise control system on dspace 35
5.9 Open loop ABS response 36
5.10 Closed loop ABS response 36
5.11 Closed loop ABS response on dspace 37
5.12 Statespace response of ABS on dspace 37
5.13 Open loop motorspeed system response 38
5.14 Closed loop motorspeed system response 38
5.15 Closed loop motor speed system response on dspace 39
5.16 Statespace response of motor speed system on dspace 39
5.17 Open loop throttle system response 40
5.18 Closed loop throttle system response 40
5.19 Closed loop throttle system response on dspace 41
5.20 Statespace response of throttle system on dspace 41

Chapter 1

Introduction

1.1 Motivation

The motivation behind this project is to develop a comprehensive understanding of the behavior of vehicle dynamics under different control systems. Our goal is to analyze and simulate the performance of these systems in real-time, providing insights into their effectiveness and potential for improvement. With the increasing complexity of modern vehicles, it is critical to have a thorough understanding of how these systems work and their impact on vehicle safety and performance.

1.2 Objective

The objective of our project is to develop a real-time simulation of vehicle dynamics and to analyze the performance and safety of various control systems used in modern automobiles. Specifically, we focus on the simulation of suspension systems, ABS systems, motor speed, throttle systems, and cruise control. We aim to compare the performance of open loop and closed loop control systems and to evaluate the effectiveness of these systems in improving the ride comfort and safety of the vehicle.

To achieve our objective, we have implemented the control systems on Microlabbox and DSPACE, which are real-time simulation platforms commonly used in the automotive industry. Through our simulation, we seek to provide insights into the design and optimization of control systems for modern automobiles and to contribute to the ongoing research and development in this field.

1.3 Problem Definition

The problem we aim to address is the lack of a comprehensive real-time simulation of vehicle dynamics that can accurately model the behavior of various control systems. While many simulations exist, few are capable of accurately modeling the complex interactions between these systems, particularly in real-time scenarios. By developing a real-time simulation, we aim to provide a more accurate and comprehensive understanding of the behavior of these systems, allowing for more effective design and optimization.

1.4 Organization of Report

The report will be structured in the following manner: The first chapter will introduce the project and cover its motivation, objectives, problem definition, and the reasons behind the project. Chapter two will discuss the literature survey. Chapter three will explain the system design, it will give an overview of vehicle dynamics, including the software and hardware used. The fourth chapter will present the system implementation that will consist of discussion on open and closed loop transfer function used to create the real-time simulation on matlab simulink, Dspace model and State Space model as well. Chapter five will discuss simulation results, outlining the performance of different control systems in various scenarios and give outcomes of the study, including an evaluation of the efficiency of the various control systems . Chapter six will conclude and summarize the primary findings of the report and provide suggestions for future research.

Chapter 2

Literature Survey

Vehicle dynamics is an essential concept within the automotive industry, holding significant importance in its advancement. In recent decades, substantial advancements have occurred in both the theoretical and practical domains of vehicle dynamics, contributing to a deeper understanding of its principles and facilitating experimental exploration.

Key components of the vehicle system include the vehicle body (sprung mass), suspension elements (spring and damper), and the tire (unsprung mass). Since the 1930s, researchers have dedicated their efforts to exploring steering mechanisms, suspension mechanics, and driving stability. Their studies have encompassed the examination of external factors such as road conditions, airflow, tires, and driver input on vehicle dynamics, as well as the interconnected relationships between these factors.

Segel provided a comprehensive overview of the advancements in vehicle dynamics up until 1990. [1] Subsequent years witnessed extensive research focused on vehicle ride comfort and handling stability. The development of vehicle handling stability underwent a transition from experimental investigations to theoretical analyses, as well as a shift from open-loop to closed-loop systems. Noteworthy contributions in the field of vehicle handling dynamics include Masato Abe's [2] "Vehicle Handling Dynamics Theory and Application" and Guo's "Vehicle Handling Dynamics Theory". [3]

The field of vehicle driving dynamics encompasses two primary categories: longitudinal dynamics and vertical dynamics. Longitudinal dynamics encompasses aspects such as driving and braking, while vertical dynamics involve ride comfort. Within the realm of vehicle dynamics, the study of longitudinal tire force has played a crucial role in addressing issues like driving slip and braking slip, ultimately leading to improvements in driving and braking efficiency.

Every vehicle that travels on uneven or randomly profiled roads is subjected to vibrations that can be detrimental to both passenger comfort and the vehicle's durability. Thus, the primary objective of a vehicle suspension system is to ensure a comfortable ride and stable road-holding capabilities across a range of road conditions and maneuvers, ultimately contributing to safety. A well-designed suspension system should provide a balance between ride comfort and handling, taking into account the specific purpose of the vehicle. It is crucial to minimize disturbances to outputs such as vehicle height in order to achieve an effective suspension design. An ideal suspension system should be able to absorb road disturbances while remaining responsive to load changes. This

paper introduces a control system for active suspension systems, which have evolved from traditional leaf springs to modern automobiles incorporating sophisticated control algorithms. The implementation involves constructing the control system's transfer function and performing simulations using MATLAB software. The results demonstrate that an active suspension system enhances ride comfort, even at resonant frequencies. For a step input of 0.08 m, the displacement of the sprung mass has been reduced by 25%, indicating an improvement in ride comfort, while the sprung mass acceleration has been reduced by 89.93%. Additionally, the suspension travel and tire deflection have been decreased by 74.64% and 89.73%, respectively. However, it is important to acknowledge that implementing an active suspension system in real-world vehicles presents challenges, including cost, space requirements, and power consumption. [14]

The performance of a vehicle's suspension system has a direct impact on various aspects of vehicle dynamics, including control stability, braking safety, ride comfort, and acceleration. Therefore, it is crucial to develop an accurate model of the suspension system in order to create a highly precise vehicle dynamics model. With this in mind, Vehicle Dynamics Analysis (ve-DYNA) offers a range of configurable suspension models, such as non-independent suspension, single-wishbone independent suspension, double-wishbone independent suspension, and double-wishbone vertical and independent suspension. In this study, we focus on the method of modeling and simulating based on ve-DYNA. We construct a vehicle dynamics model for a specific vehicle type and develop a Hardware-in-the-Loop Simulation (HILS) system for a particular vehicle control unit. By leveraging ve-DYNA and our developed HILS system, we aim to enhance our understanding and analysis of vehicle dynamics while providing a valuable tool for testing and validating vehicle control units. [15]

Steering and braking control systems are crucial for preventing rollover incidents in road vehicles. The proposed control concept consists of three feedback loops: continuous operation steering control, emergency steering control, and emergency braking control. In continuous operation, the roll rate and roll acceleration are utilized as feedback signals, with velocity scheduled gains applied to the front wheel steering angle. This approach robustly enhances the vehicle's roll damping across a wide range of speeds and center of gravity heights. It is important to consider that the center of gravity may vary, especially in vehicles like trucks where it can change from one ride to another. This work presents a control concept that extends the linear steering control by incorporating nonlinear emergency steering and braking control strategies. By integrating these components, the overall control system aims to provide enhanced safety and stability in critical situations. [16]

The automobile cruise control system serves as an automatic speed control mechanism, allowing the car to maintain a consistent speed during a journey. The system's output, which is the speed, is regulated by a controller to achieve the desired speed set by the driver. Traditionally, drivers have to continuously press the acceleration pedal to maintain the car's speed. However, the cruise control system provides convenience and comfort by eliminating the need for frequent pedal adjustments, reducing driver fatigue. It offers an easier way to control the car's speed through buttons instead of pedals. This study focuses on evaluating the performance of various control approaches, including conventional, modern, and intelligent controllers, implemented in a cruise control system. The cruise control system is a popular and valuable model in control system engineering

education, as it provides a simple yet comprehensive platform for exploring classical and modern control design methods. In this research, mathematical models are developed for both linear and nonlinear dynamic models of the cruise control system. Linear models are controlled using PID, state space, and artificial intelligence controllers (such as fuzzy logic), while a PID controller with feedforward is proposed for the nonlinear model with disturbance effects. Furthermore, a feedforward-type PD controller is introduced to mitigate gravitational and wind disturbances. [4]

In this study, a throttle/brake control law specifically designed for the intelligent cruise control (ICC) system is proposed. The ICC system comprises a vehicle detection sensor, a controller, and throttle/brake actuators. To control the throttle/brake system, two components are introduced: a solenoid-valve-controlled electronic vacuum booster (EVB) and a step-motor-controlled throttle actuator. A nonlinear computer model is developed for the electronic vacuum booster, and simulations are conducted using a comprehensive nonlinear vehicle model. The proposed control law includes an algorithm that generates the desired acceleration/deceleration profile in ICC scenarios, a throttle/brake switching logic, and a throttle and brake control algorithm based on vehicle dynamics. To evaluate the control performance, computer simulations, and experiments are conducted. The findings demonstrate that the proposed ICC throttle/brake control law delivers satisfactory vehicle-to-vehicle distance and velocity control performance. [5] [10]

This study focuses on the vehicle dynamics control system designed for a four-in-wheel motor drive electric vehicle, with the objective of enhancing vehicle stability during challenging driving situations. The vehicle dynamics controller comprises three modules: motion following control, control allocation, and vehicle state estimation. To address the nonlinear behavior of tires under critical driving conditions, the yaw motion of the vehicle is regulated using gain scheduling control based on the linear quadratic regulator theory. The feed-forward and feedback gains of the controller are continuously updated in real time by estimating the tire cornering stiffness. This approach ensures control robustness against environmental disturbances and parameter uncertainties. The control allocation module assigns the calculated generalized force requirements to each in-wheel motor, taking into account the tire longitudinal/lateral force coupling characteristic, utilizing quadratic programming theory. Simulations are conducted under various driving conditions to validate the effectiveness of the control algorithm. The simulation results demonstrate that the proposed vehicle stability controller successfully stabilizes the vehicle motion during critical driving conditions. [6]

Chapter 3

System Design

3.1 Overview of vehicle dynamics

The study of vehicle dynamics is crucial for ensuring the safety and efficiency of modern vehicles. Advanced control systems have made it necessary to examine and simulate the behavior of these systems in real-time. This project focuses on creating a real-time simulation of vehicle dynamics, with a special emphasis on control systems such as suspension, ABS, motor speed, throttle, and cruise control. The main objective of this project is to gain a complete understanding of how these systems behave under different conditions and to analyze their effectiveness and potential for improvement. It is essential to comprehend the workings of these systems and their impact on vehicle performance and safety due to the increasing complexity of modern vehicles.

Overview of the different components involved in vehicle dynamics: Vehicle dynamics involve the study of various components that work together to produce the vehicle's overall behavior. The main components involved in vehicle dynamics include the sprung mass, suspension system, and tires.

Sprung Mass: The sprung mass refers to the total weight of the vehicle, including the body and all the components mounted on the body, such as the engine, transmission, and occupants. The sprung mass is supported by the suspension system and influences the vehicle's ride comfort and handling.

Suspension System : The suspension system consists of springs and dampers that are designed to absorb and dampen the forces generated by the vehicle's motion. The suspension system not only supports the sprung mass but also helps maintain tire contact with the road surface, which is crucial for handling and stability.

Tires: The tires are the only point of contact between the vehicle and the road surface. The tire's characteristics, such as size, shape, and composition, play a significant role in the vehicle's performance and safety. Tires also affect the vehicle's ride comfort and handling, and they can have a significant impact on fuel economy. All three components work together to influence the vehicle's behavior.

3.2 Hardware Used

The hardware design of our system involves the selection and configuration of the microlabbox - dSPACE system and Digital Storage Oscilloscope (DSO) to simulate the different control systems of the vehicle dynamics. The microlabbox and dSPACE systems are chosen because they are widely used in the automotive industry and provide a high level of accuracy and reliability. The DSO is used to measure and analyze the signals generated by the microlabbox and dSPACE systems. All the hardware components are connected to each other via Ethernet cables and are also connected to the computer running the MATLAB software.

3.2.1 Microlabbox

The Microlabbox is a compact, real-time system that is commonly used for rapid control prototyping and hardware-in-the-loop simulations. It is a modular system that allows for the integration of different I/O modules and can be configured for a variety of applications. The Microlabbox is equipped with a powerful processor and a FPGA for high-speed signal processing. It also has Ethernet connectivity for communication with external devices and a range of software tools for simulation, modeling, and code generation.

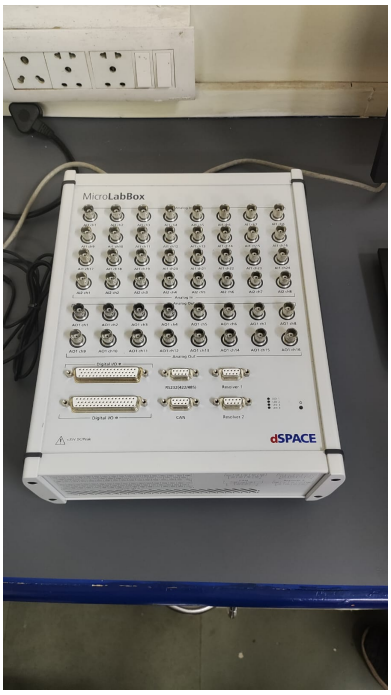


Figure 3.1: Dspace Microlabbox

3.2.2 Digital Storage Oscilloscope

The Digital Storage Oscilloscope (DSO) is a measuring instrument that is used to visualize and analyze electrical signals. It can capture and store signals in its memory for later analysis, and it has the ability to trigger on specific events or signal patterns. The DSO can also perform advanced signal analysis functions such as Fast Fourier Transform(FFT) and waveform math operations.

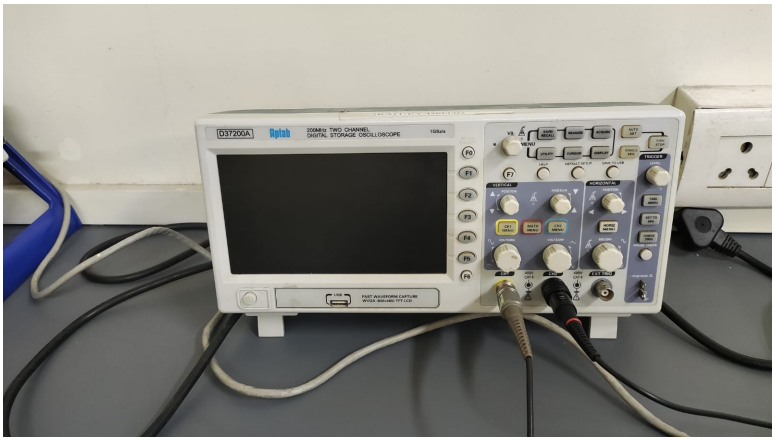


Figure 3.2: Digital Storage Oscilloscope

3.3 Software Used

3.3.1 Matlab

MATLAB (short for "matrix laboratory") is a programming language and numerical computing environment developed by MathWorks. It was first released in 1984 and has since become one of the most popular tools for scientific computing and data analysis. MATLAB provides a wide range of built-in functions for mathematical operations, data analysis, and visualization. It also supports the development of user-defined functions and toolboxes, making it a flexible and powerful tool for solving complex mathematical problems.

Some of the key features of MATLAB include:

- An intuitive and interactive environment for data analysis and visualization
- A wide range of built-in functions for mathematical operations, statistics, optimization, and signal processing
- Support for user-defined functions and toolboxes for specific applications
- Integration with other programming languages, such as C/C++, Java, and Python
- Support for parallel computing and GPU acceleration for faster computation
- Availability of MATLAB Online, a cloud-based version of MATLAB for web-based access and collaboration

MATLAB Simulink

MATLAB Simulink is a block diagram environment for modeling, simulating, and analyzing dynamic systems. It is built on top of MATLAB and provides a graphical user interface for designing complex models and simulating the behavior of the system under different conditions. Simulink models can be used to simulate a wide range of systems, including control systems, communication systems, signal processing systems, and power systems.

- An intuitive and interactive environment for designing and simulating complex systems
- A wide range of built-in blocks for modeling complex systems, including linear and nonlinear systems, continuous-time and discrete-time systems, and hybrid systems
- Support for system-level design, including automatic code generation and verification
- Integration with MATLAB for advanced analysis and visualization of simulation results
- Availability of Simulink Online, a cloud-based version of Simulink for web-based access and collaboration

Chapter 4

System Implementation

System implementation involves the development and testing of various control systems for vehicle dynamics using a combination of simulation techniques. We have simulated the performance of these control systems using open and closed loop diagrams and have developed state-space models of transfer functions for each system. We have also implemented each transfer function on dspace.

To test the performance of these control systems in a real-world environment, we have implemented them on the Microlabbox and DSPACE platforms, which provide real-time simulation. We have used these platforms to verify the effectiveness of our control systems in improving the ride comfort and safety of the vehicle, and to identify any issues or limitations that may arise during operation.

4.1 Suspension System

The suspension system is an essential component of any vehicle that provides a smooth and comfortable ride to the passengers. It is responsible for absorbing the impact of road irregularities and disturbances and transmitting them to the vehicle chassis and body, minimizing the effects of vibrations and shocks on the passengers and cargo.

In our project on vehicle dynamics, we have developed a simulation of a suspension system and evaluated its performance using open and closed loop control. The suspension system is modeled as a multi-degree-of-freedom system, where each wheel of the vehicle is connected to a spring and a damper. The spring provides the restoring force that opposes the deformation of the wheel due to road irregularities, while the damper dissipates the energy of the oscillation and prevents the wheel from bouncing.

Our analysis of the results shows that the active suspension system provides significant improvements in ride comfort and handling compared to the passive suspension system. The active suspension system is better able to adapt to changing driving conditions and disturbances, providing a smoother and more stable ride to the passengers.

Transfer Function Equation

$$G = \frac{2820s^2 + 15020s + 500000}{800000s^4 + 3.854 \times e^07s^3 + 1.481 \times e^09s^2 + 1.377 \times e^09s + 4 \times e^{10}} \tag{4.1}$$

Open Loop Block Diagram

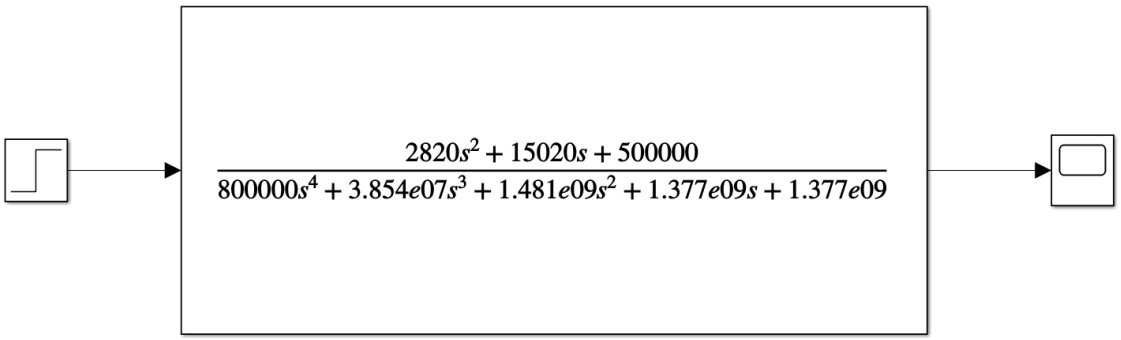


Figure 4.1: Open Loop Suspension System

Closed Loop Block Diagram

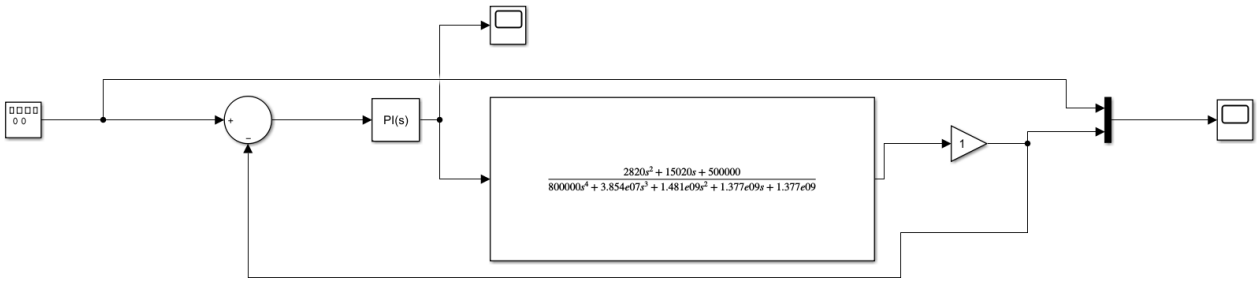


Figure 4.2: Closed Loop Suspension System

Dspace Model

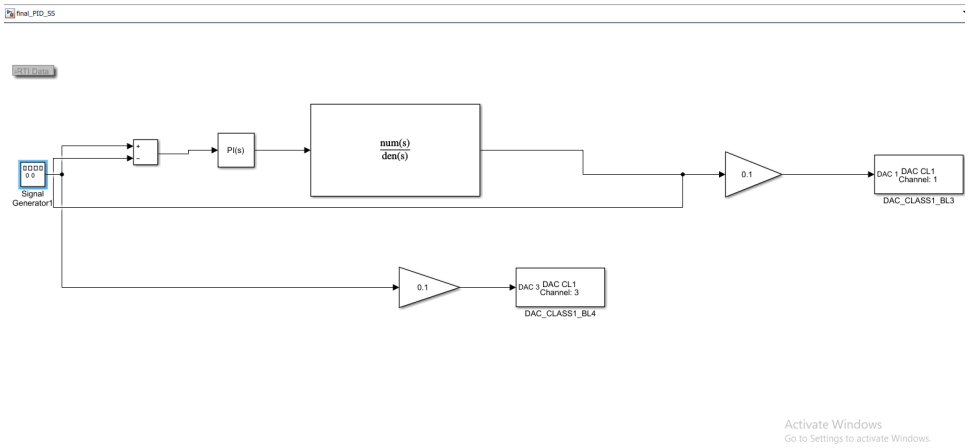


Figure 4.3: Dspace model of Suspension System

In order to achieve optimal results in terms of settling time, we meticulously adjusted the specific parameters of P, I, and D. After thorough analysis, it was determined that a PI controller was the most suitable for suspension system, offering both robustness and a faster settling time. Consequently, we concluded by implementing a PI controller for suspension system, utilizing the parameter values specified in the accompanying figure.

PID block parameters

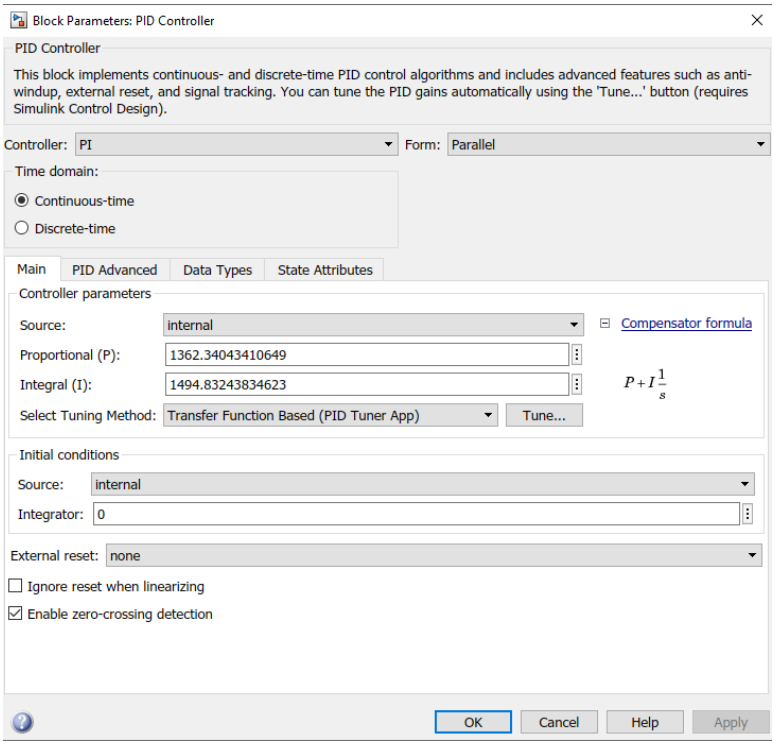


Figure 4.4: PID parameter of Suspension System

State Space Model

In order to examine the behavior of the suspension system over time, we developed a block diagram with predefined states that incorporates variable feedback. Since the transfer function of the suspension system has a fourth-order, the block parameters include four state variables.

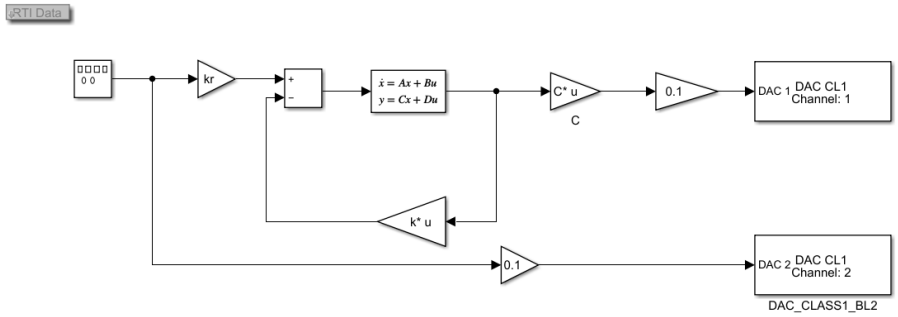


Figure 4.5: State Space Model

Block parameter of State Space Model

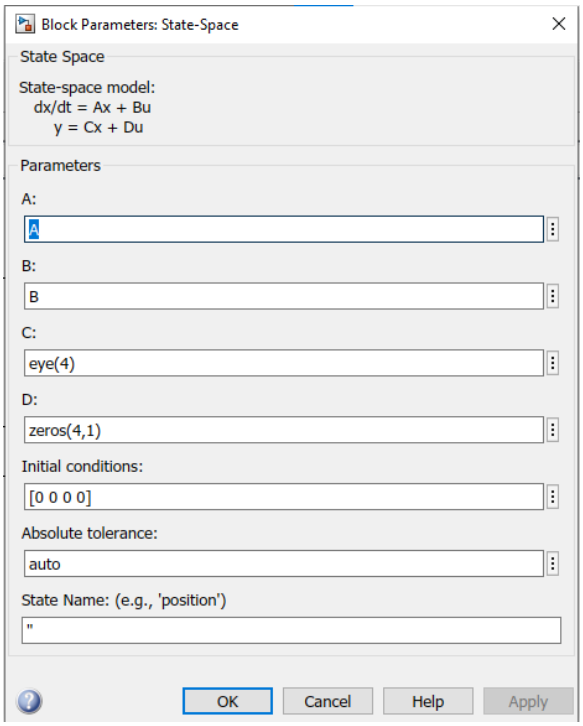


Figure 4.6: Suspension system - Statespace model block parameter

4.2 Cruise Control System

Cruise control is a system used in automobiles that allows the driver to maintain a constant speed without the need for constant acceleration or deceleration. The system uses sensors and electronic control systems to automatically adjust the vehicle's throttle or engine power to maintain a steady speed, providing a more comfortable and fuel-efficient driving experience.

In our project on vehicle dynamics, we have developed a simulation of a cruise control system and evaluated its performance using open and closed loop control. The cruise control system is modeled as a feedback control system, where the vehicle's speed is compared to a set reference speed, and the throttle or engine power is adjusted to maintain the desired speed.

We have implemented the cruise control system on the Microlabbox and DSPACE platforms to evaluate its performance in various driving scenarios. Our analysis of the results shows that the cruise control system is effective in maintaining a constant speed and improving fuel efficiency by reducing the need for constant acceleration and deceleration. However, the performance of the system can be affected by external disturbances, such as wind, hills, and traffic, which can cause fluctuations in the vehicle's speed.

Transfer Function Equation

$$G = \frac{1}{1000s + 50} \tag{4.2}$$

Open Loop Block Diagram

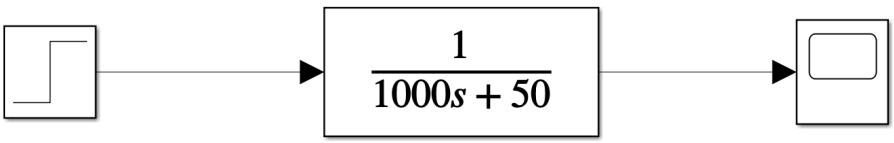


Figure 4.7: Open Loop Cruise Control System

Closed Loop Block Diagram

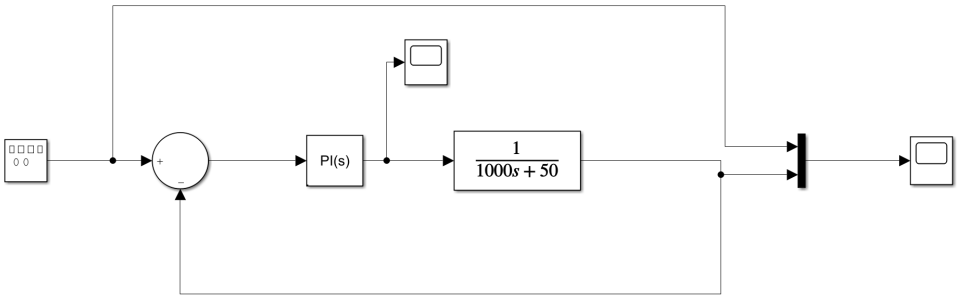


Figure 4.8: Closed Loop Cruise Control System

Dspace Model

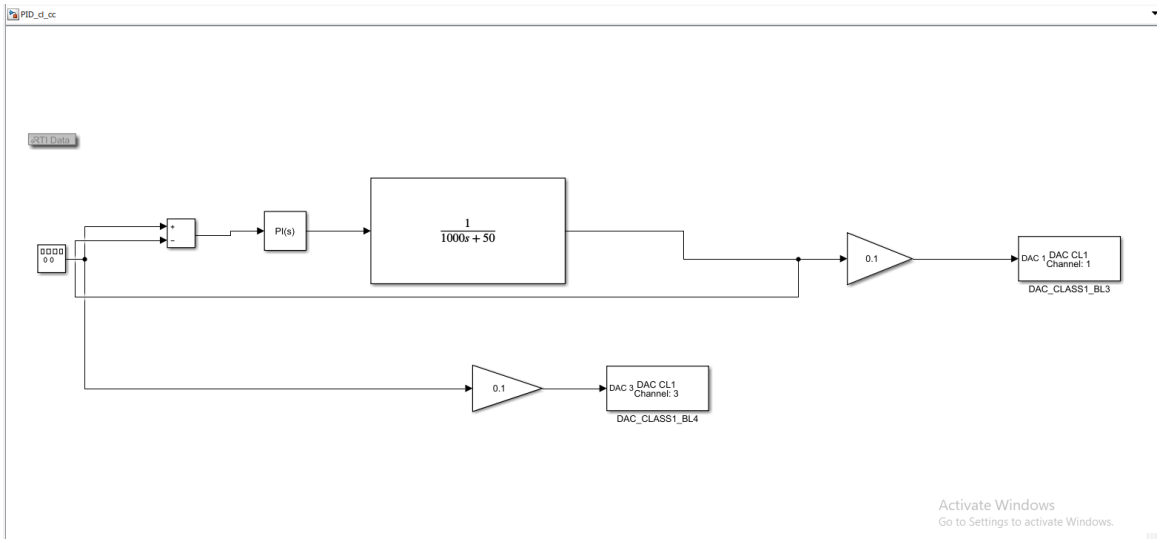


Figure 4.9: Dspace model of Cruise Control System

In order to achieve optimal results in terms of settling time, we meticulously adjusted the specific parameters of P, I, and D. After thorough analysis, it was determined that a PI controller was the most suitable for cruise control system, offering both robustness and a faster settling time. Consequently, we concluded by implementing a PI controller for cruise control system, utilizing the parameter values specified in the accompanying figure.

PID block parameters of cruise control system

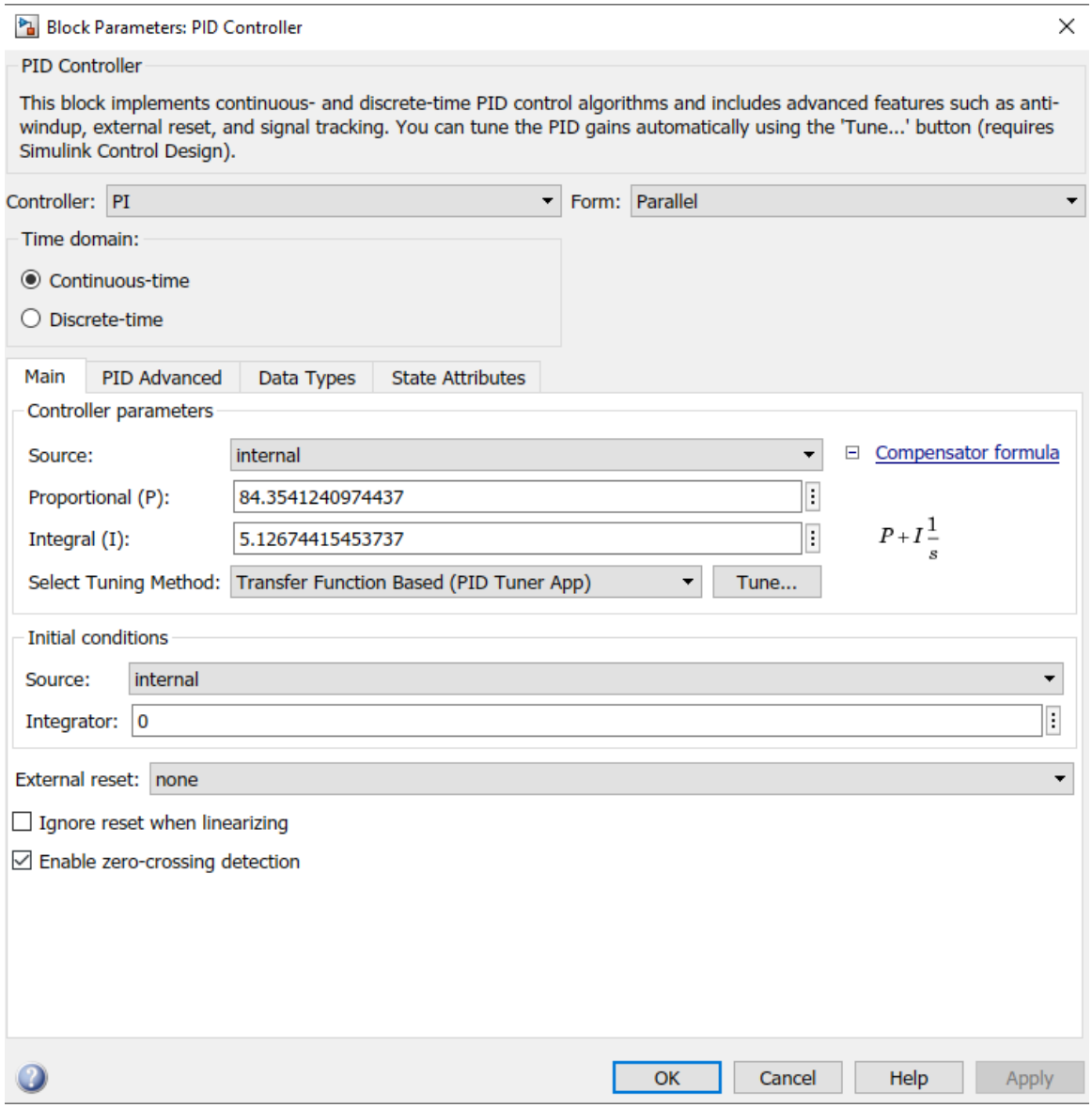


Figure 4.10: PID parameter of Cruise control System

State Space Model

In order to examine the behavior of the cruise control system over time, we developed a block diagram with predefined states that incorporates variable feedback. Since the transfer function of the cruise control system has a first-order, the block parameters include one state variable.

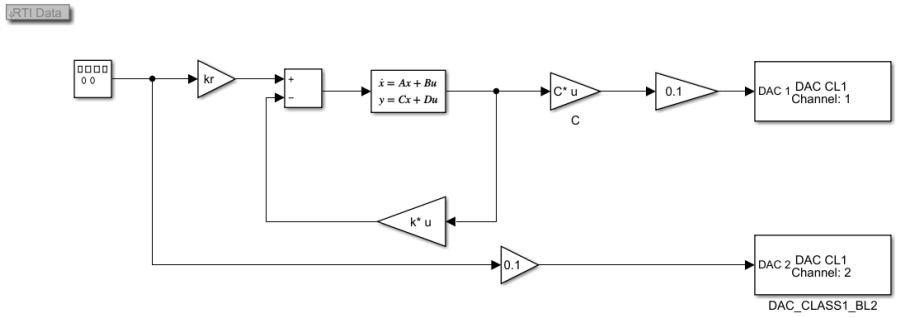


Figure 4.11: State Space Model

Block parameter of State Space Model

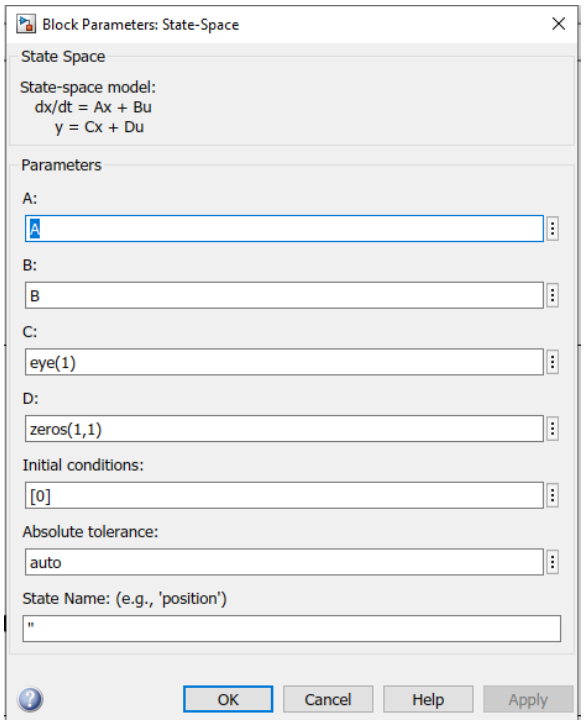


Figure 4.12: Cruise control system -Statespace model block parameter

4.3 Antilock Braking System

An Antilock Braking System (ABS) is a safety feature used in modern automobiles to prevent the wheels from locking up during braking, thereby helping to maintain steering control and prevent skidding. The system uses sensors and electronic control systems to monitor the rotational speed of the wheels and modulate the brake pressure accordingly.

When the brakes are applied, the ABS system detects if any wheel is about to lock up due to excessive braking force, which could lead to skidding and loss of control. The system then modulates the brake pressure on that wheel to reduce the braking force, preventing it from locking up while still maintaining braking effectiveness.

The ABS system operates by rapidly cycling the brake pressure, releasing it for a short duration and then reapplying it in a controlled manner, which allows the wheel to continue rotating and maintain traction. This cycling of the brake pressure is performed many times per second, providing a smoother and more controlled braking experience.

ABS systems have been shown to significantly improve vehicle safety, particularly in adverse driving conditions such as wet or icy roads, where skidding and loss of control are more likely. ABS is now a standard safety feature in most modern vehicles, and many countries have made it mandatory for all new vehicles to be equipped with ABS.

In our project on vehicle dynamics, we have developed a simulation of an ABS system and evaluated its performance using open and closed loop control. The simulation includes the modeling of the vehicle dynamics and the ABS system, which allows us to test the system's performance in various driving scenarios.

Our analysis of the results shows that the ABS system is effective in preventing wheel lockup and improving vehicle stability and control during braking. However, the performance of the system can be affected by external factors such as road surface conditions and tire characteristics. Therefore, it is important to design the ABS system with these factors in mind and to test its performance under realistic driving conditions.

Transfer Function Equation

$$G = \frac{10}{s^2 + 0.7s + 0.1} \tag{4.3}$$

Open Loop Block Diagram

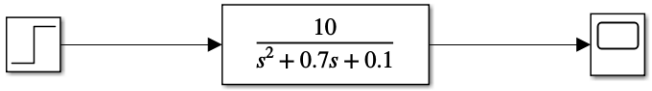


Figure 4.13: Open Loop Antilock Braking System

Closed Loop Block Diagram

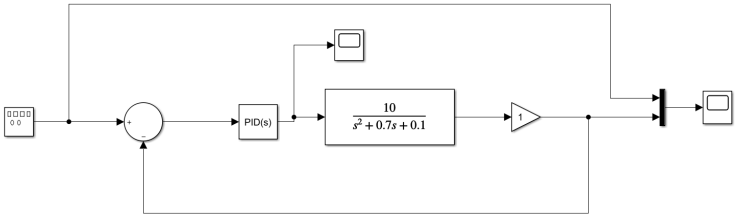


Figure 4.14: Closed Loop Antilock Braking System

Dspace Model

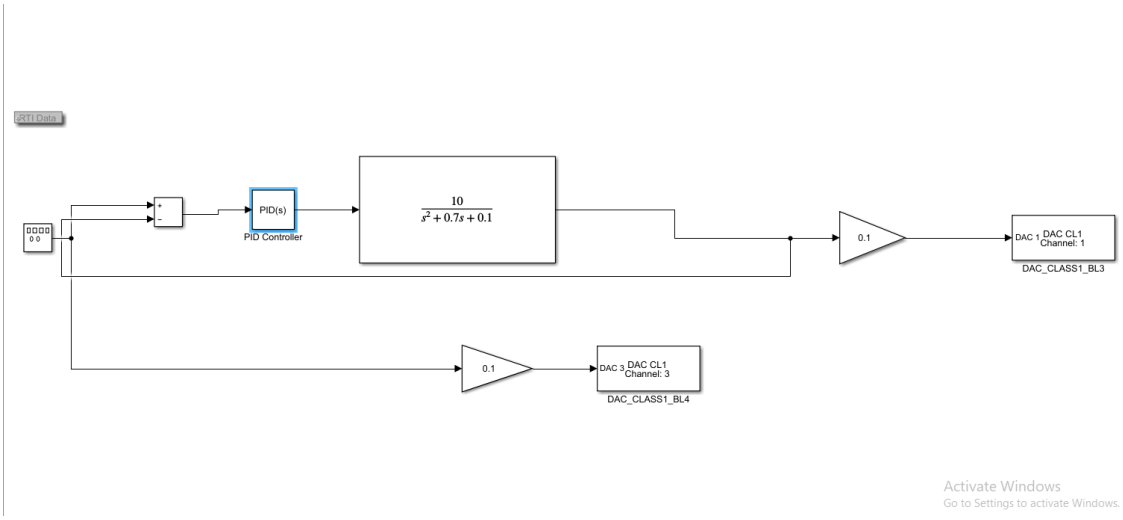


Figure 4.15: Dspace model of ABS System

In order to achieve optimal results in terms of settling time, we meticulously adjusted the specific parameters of P, I, and D. After thorough analysis, it was determined that a PID controller was the most suitable for ABS, offering robustness, reduced overshoot and a faster settling time. Consequently, we concluded by implementing a PID controller for ABS, utilizing the parameter values specified in the accompanying figure.

PID block parameters of ABS system

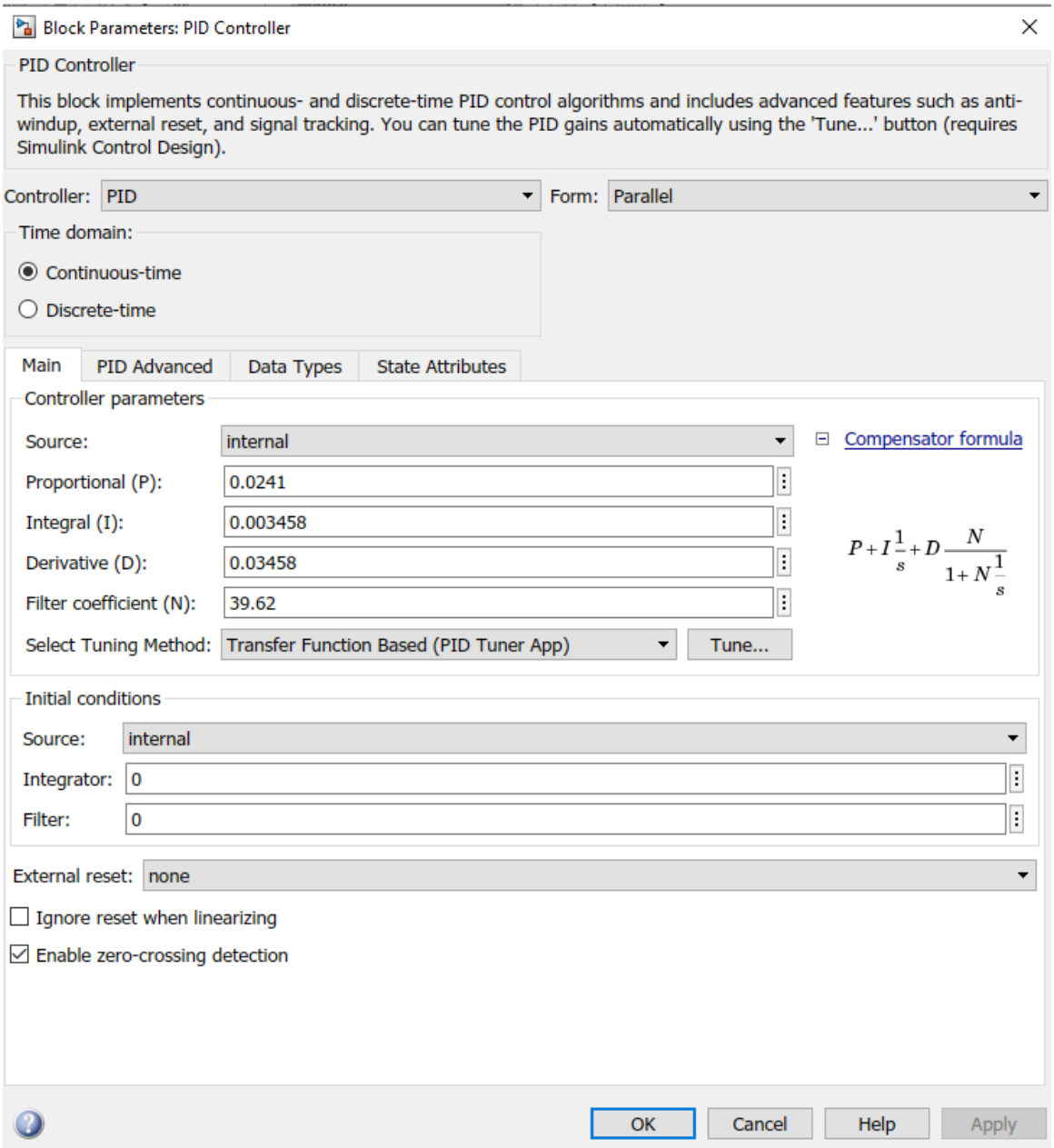


Figure 4.16: PID parameter of anti lock breaking System

State Space Model

In order to examine the behavior of the antilock braking system over time, we developed a block diagram with predefined states that incorporates variable feedback. Since the transfer function of the antilock braking system has a second-order, the block parameters include two state variables.

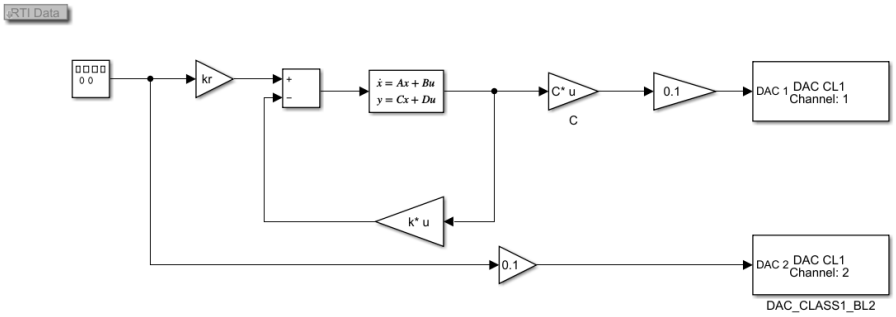


Figure 4.17: State Space Model

Block parameter of State Space Model

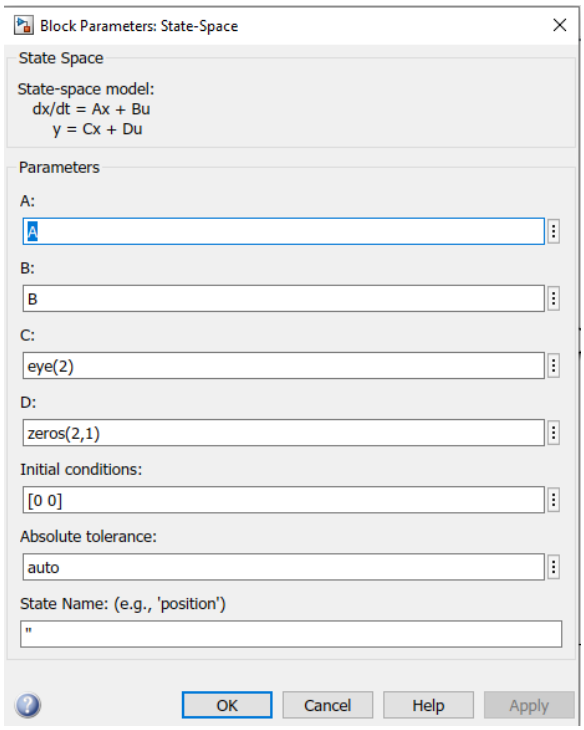


Figure 4.18: ABS - Statespace model block parameter

4.4 Motor Speed System

A motor speed system is a control system used to regulate the speed of an electric motor. In our project on vehicle dynamics, we have developed a simulation of a motor speed system to analyze its performance under different conditions.

The motor speed system typically involves a feedback loop that continuously monitors the motor speed and adjusts the input voltage to the motor to maintain a constant speed. The feedback loop consists of a speed sensor, which measures the actual motor speed, and a controller, which compares the actual speed to a desired speed and adjusts the input voltage accordingly.

The controller in the motor speed system can be either a simple proportional-integral-derivative (PID) controller or a more complex model-based controller, depending on the requirements of the application.

In our simulation, we have modeled the motor speed system using a open loop, closed loop and state-space representation and implemented it using MATLAB/Simulink.

Our analysis of the results shows that the motor speed system is effective in maintaining a constant motor speed under different operating conditions. However, the performance of the system can be affected by factors such as motor temperature, mechanical wear, and voltage fluctuations. Therefore, it is important to design the system with these factors in mind and to continuously monitor and maintain the system to ensure optimal performance.

Transfer Function Equation

$$G = \frac{0.01}{0.005s^2 + 0.06s + 0.1001} \tag{4.4}$$

Open Loop Block Diagram

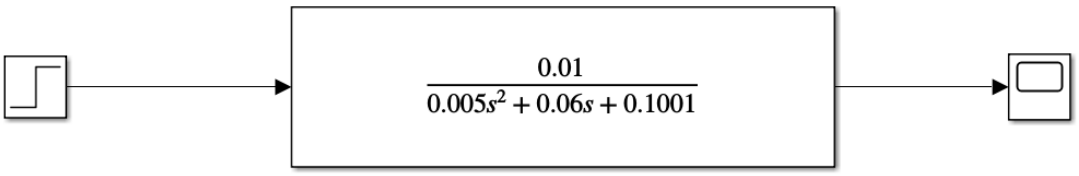


Figure 4.19: Open Loop Motor Speed System

Closed Loop Block Diagram

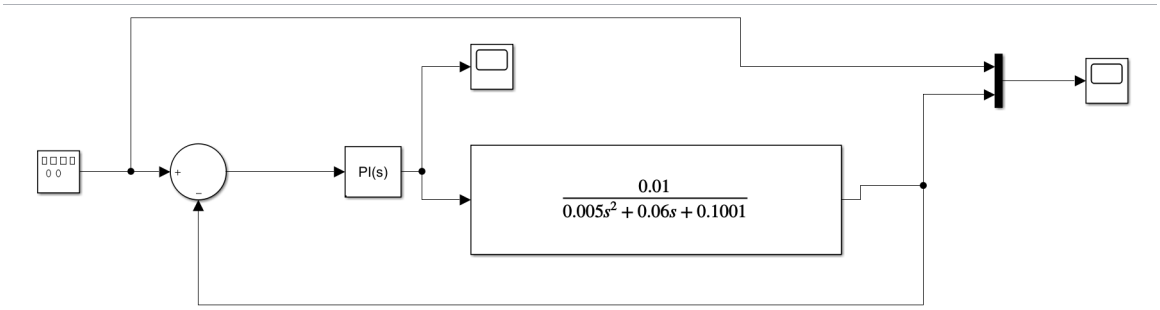


Figure 4.20: Closed Loop Motor Speed System

Dspace Model

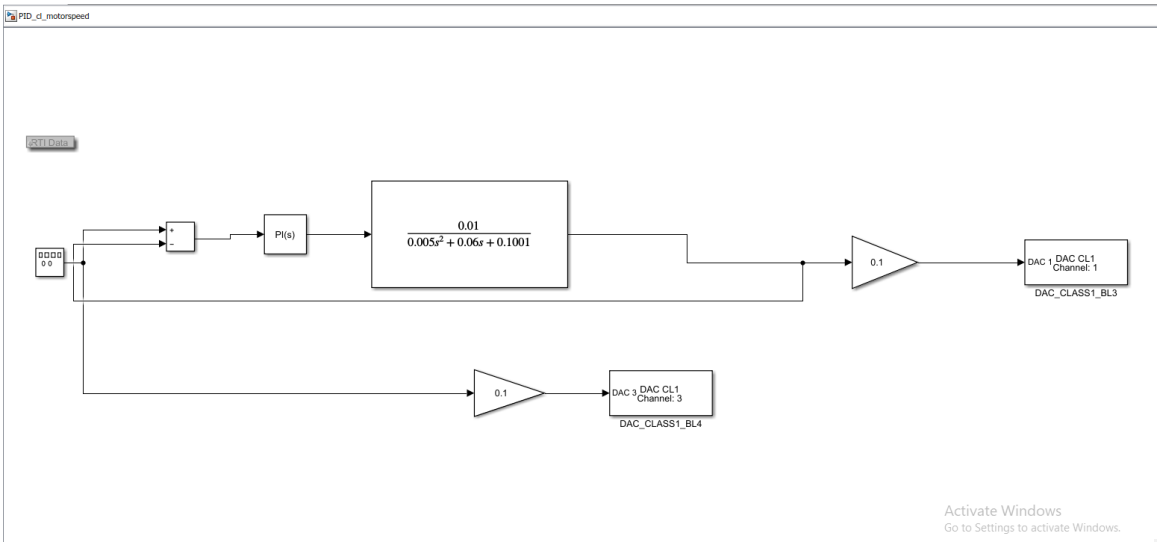


Figure 4.21: Dspace model of motor speed System

In order to achieve optimal results in terms of settling time, we meticulously adjusted the specific parameters of P, I, and D. After thorough analysis, it was determined that a PI controller was the most suitable for motor speed system, offering both robustness and a faster settling time. Consequently, we concluded by implementing a PI controller for motor speed system, utilizing the parameter values specified in the accompanying figure.

PID block parameters of motor speed system

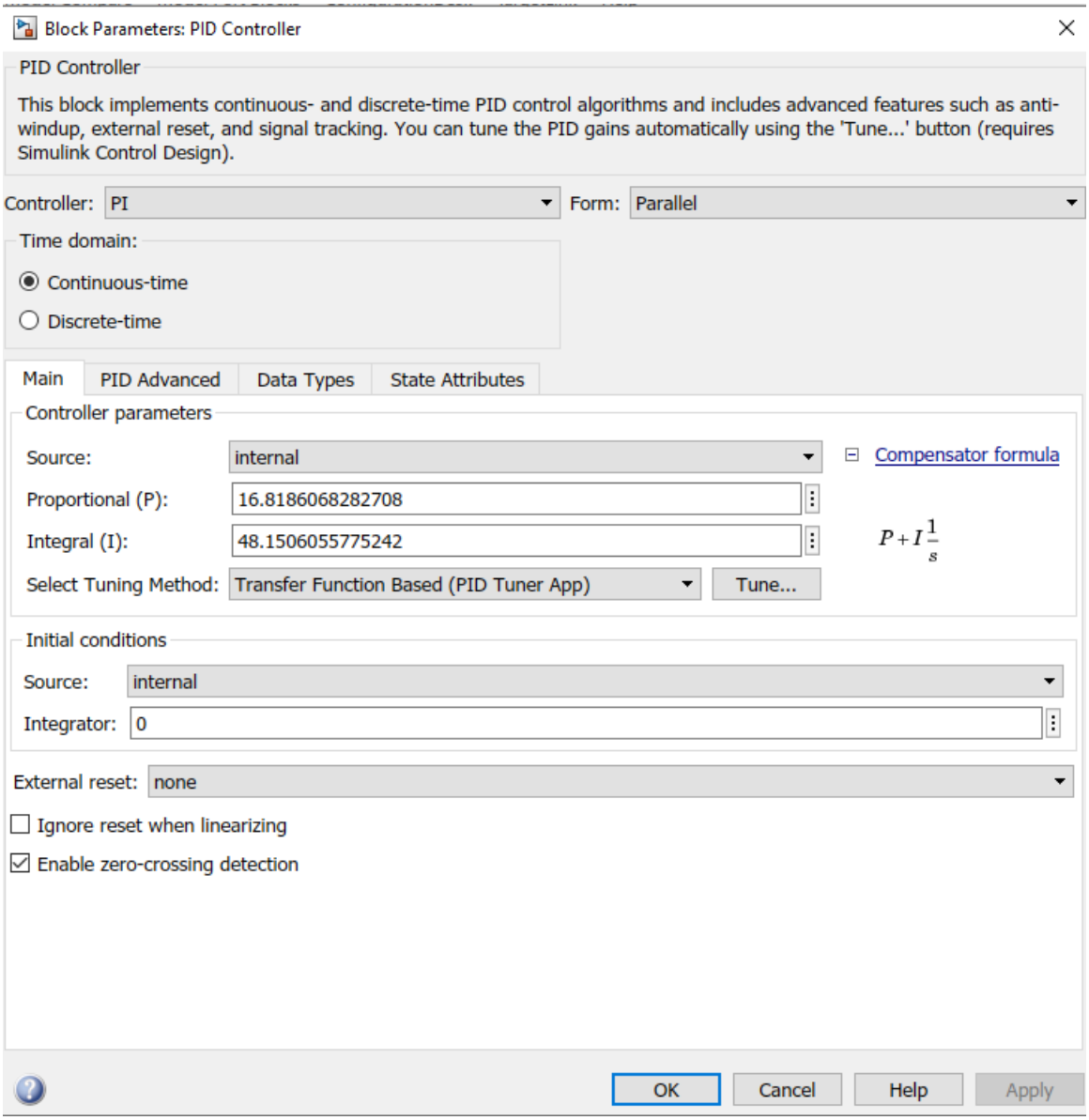


Figure 4.22: PID parameter of motor speed system

State Space Model

In order to examine the behavior of the motor speed system over time, we developed a block diagram with predefined states that incorporates variable feedback. Since the transfer function of the motor speed system has a second-order, the block parameters include two state variables.

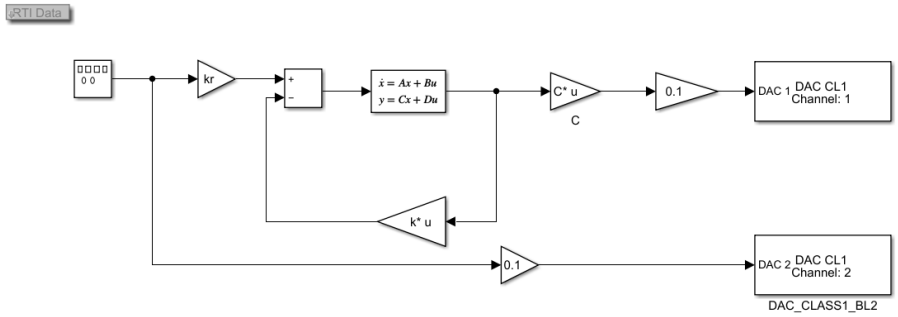


Figure 4.23: State Space Model

Block parameter of State Space Model

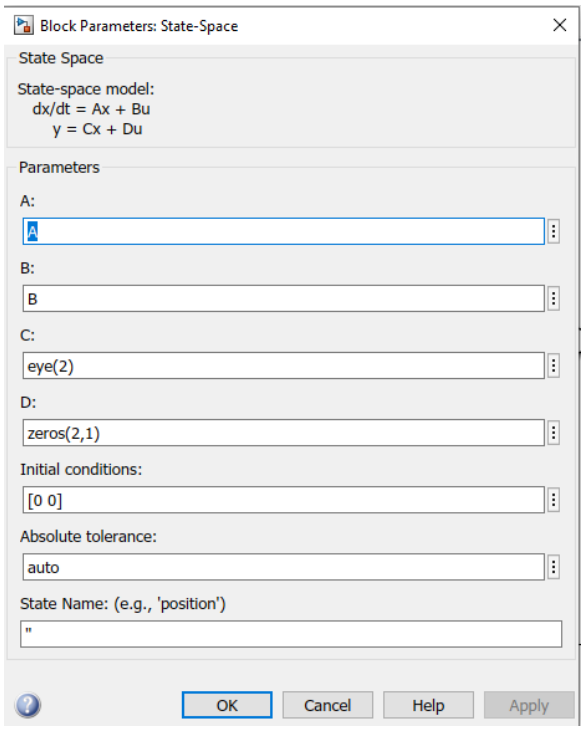


Figure 4.24: Motor speed - Statespace model block parameter

4.5 Throttle System

The throttle system is a critical control system in a vehicle that regulates the amount of air and fuel entering the engine's combustion chamber. The accelerator pedal controls the throttle valve in the engine's air intake system, which adjusts the amount of air that flows into the engine. The fuel injector then injects the appropriate amount of fuel into the engine's combustion chamber, resulting in combustion and the generation of power.

The throttle system consists of several components, including the throttle pedal, throttle cable, throttle body, throttle valve, and throttle position sensor. The throttle position sensor measures the position of the throttle valve and sends signals to the engine control unit (ECU) to adjust the fuel injection and ignition timing accordingly.

In modern vehicles, electronic throttle control systems (ETCS) have replaced traditional mechanical throttle systems. The ETCS uses an electronic throttle actuator to control the throttle valve position, which is managed by the ECU. The use of ETCS improves vehicle performance, fuel efficiency, and reduces emissions.

In our project, we have developed a simulation model of the throttle system and analyzed its performance in both open-loop and closed-loop control systems. We have implemented the throttle system control on microlab box and dSpace, and analyzed the results. Our analysis and simulation results provide insights into the performance and behavior of the throttle system under different conditions, which can be useful for the development and optimization of throttle control strategies in vehicles.

Transfer Function Equation

$$G(s) = \frac{1}{s^2 + s + 1.1} \tag{4.5}$$

Open Loop Block Diagram

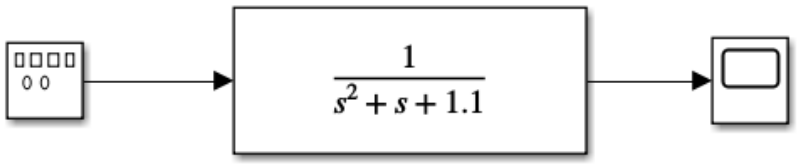


Figure 4.25: Open Loop Throttle System

Closed Loop Block Diagram

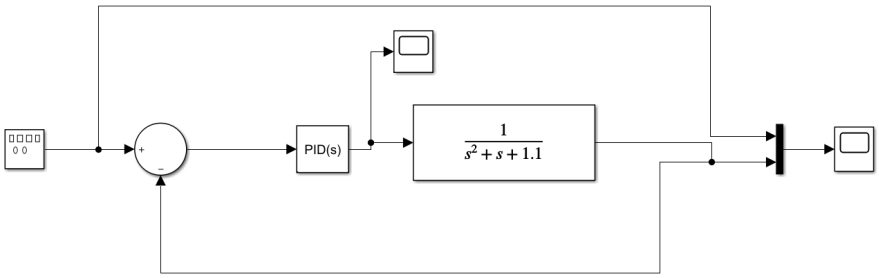


Figure 4.26: Closed Loop Throttle System

Dspace Model

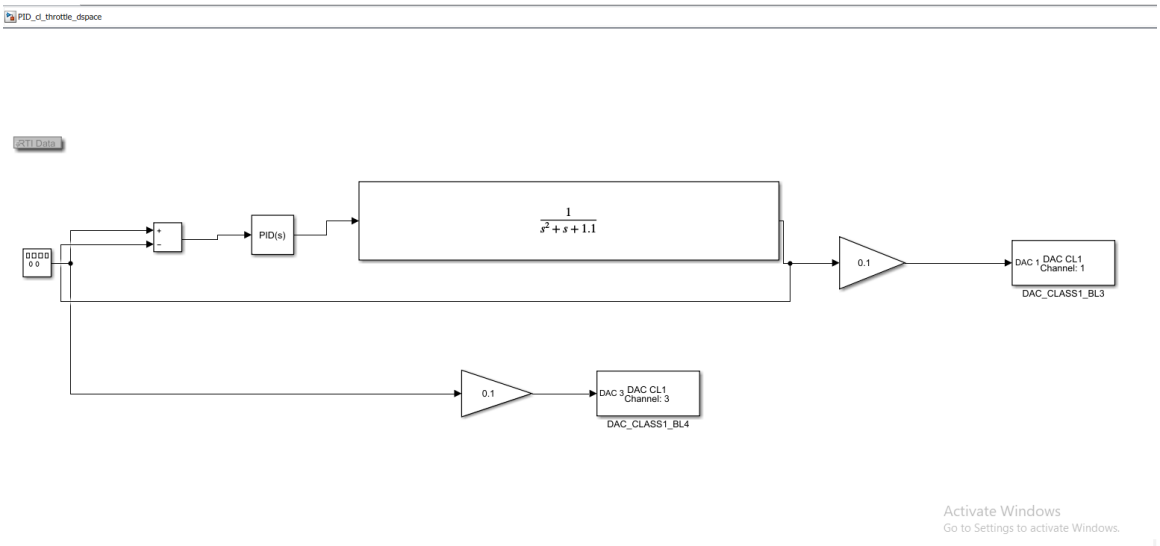


Figure 4.27: Dspace model of Throttle System

In order to achieve optimal results in terms of settling time, we meticulously adjusted the specific parameters of P, I, and D. After thorough analysis, it was determined that a PID controller was the most suitable for throttle system, offering reduced overshoot, robustness and a faster settling time. Consequently, we concluded by implementing a PID controller for throttle system, utilizing the parameter values specified in the accompanying figure.

PID block parameters of throttle system

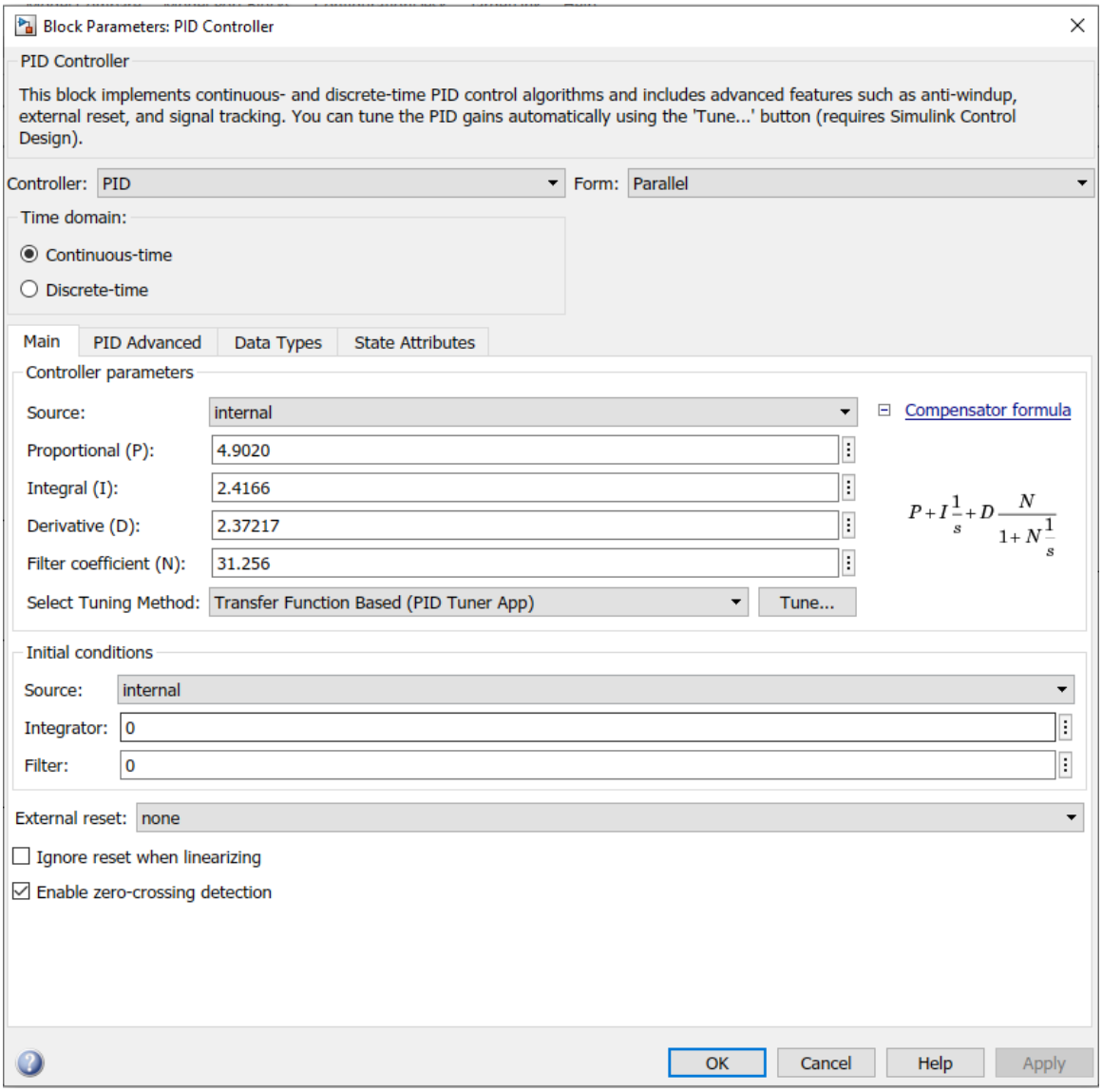


Figure 4.28: PID parameter of throttle system

State Space Model

In order to examine the behavior of the throttle system over time, we developed a block diagram with predefined states that incorporates variable feedback. Since the transfer function of the throttle system has a second-order, the block parameters include two state variables.

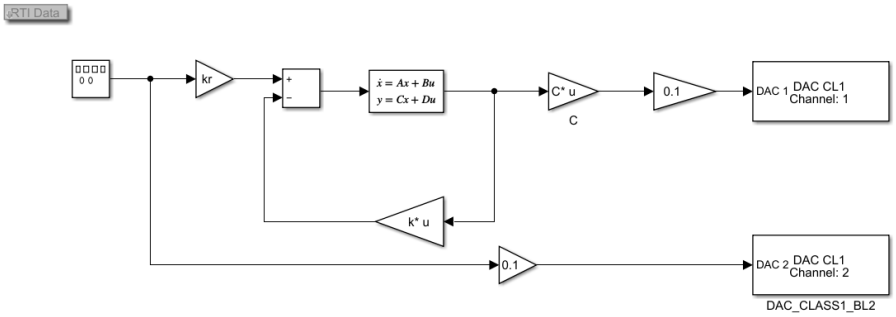


Figure 4.29: State Space Model

Block parameter of State Space Model

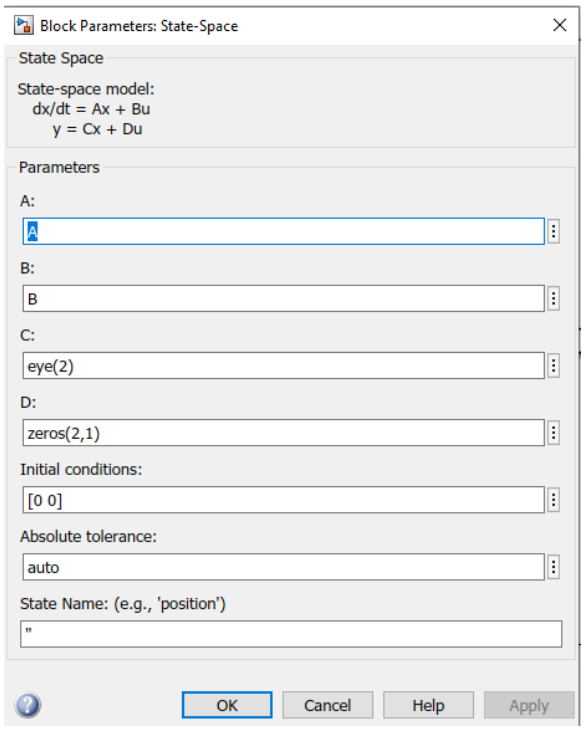


Figure 4.30: Throttle system - Statespace model block parameter

Chapter 5

Result and Analysis

Using MATLAB Simulink model we have designed and implemented the transfer function of various systems of vehicle like suspension, cruise control, antilock braking system, motor speed, and throttle system. The results for these transfer functions are taken using the Simulink. The response of both the open loop as well as the closed loop is observed and analyzed based on which conclusion is made. Further, for hardware implementation, we have used Dspace Microlabbox and Dspace. To fulfill the aim of the project which is a real-time implementation of the transfer function, the transfer function is implemented on Dspace. This helps in analyzing complex results in real time thus providing us with an efficient method. To see the results, all of the files, including the main file, system files, and packages, are imported from MATLAB.

5.1 Suspension System

A bus suspension system that is considered good should be able to maintain its stability on the road while also ensuring that passengers are comfortable during bumpy rides. Moreover, when the bus encounters any road disturbance such as potholes, cracks, or uneven pavement, the bus body shouldn't sway excessively, and any oscillations should settle down promptly. To test the suspension system's response to road disturbances, a simulated step input is used, which represents the bus coming out of a pothole. Refer 4.1 for open loop block diagram.

Suspension System Open Loop Response:

From the graph of the open-loop response for a unit step actuated force, we can see that the system is under-damped. Under-damped systems exhibit oscillatory behavior in their response, which means that the system will oscillate around the equilibrium state before eventually settling down. The complex roots indicate the presence of oscillations in the system's response. Here as seen from the response people sitting in the bus will feel very small amount of oscillation. Moreover, the bus takes an unacceptably long time to reach the steady state (the settling time is very large). The oscillation caused by the large overshoot and long settling time can make the passengers sitting in the bus uncomfortable. To improve the system's performance, a feedback controller can be added. The following is the schematic of the closed-loop system.

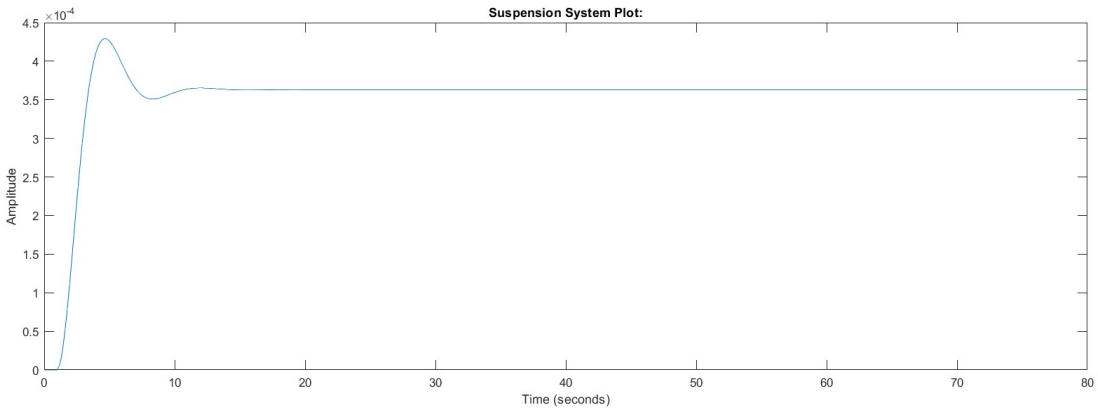


Figure 5.1: Open loop suspension system response

Suspension System Close Loop Response:

We have built a plant with PI controller (Refer 4.2 closed loop block diagram). Higher value of P can help reduce the settling time, while higher values of I can help in reducing steady-state errors. We have chose such values of P and I which has reduced the overshoot which we can see from the closed loop response, that the amplitude of oscillation is reduced with less settling time. In the following graph, blue is input and red is output response.

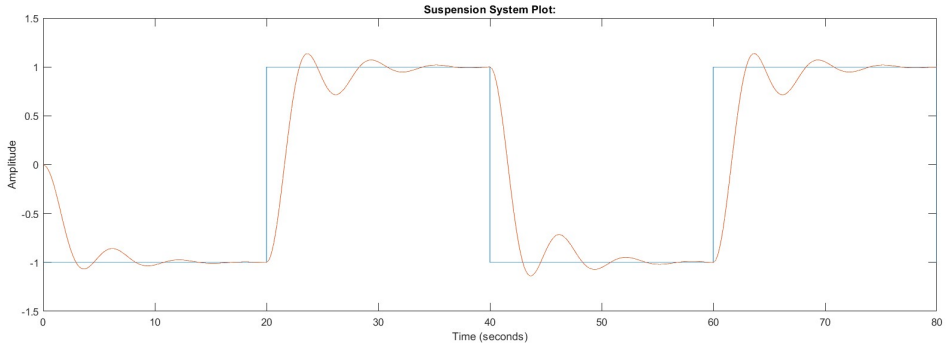


Figure 5.2: Closed loop suspension system response

Further, the real-time implementation of the closed loop suspension system is done on dspace.

After successfully testing and applying the open loop and closed loop features of the individual transfer functions on Dspace and analyzing their frequency domain responses, the same process was repeated for the state-space model of the transfer function in the time domain. In order to examine the characteristics of the transfer function in the state-space, the strategy of pole placement was employed to ensure stability in the time domain.

Closed loop dspace response:

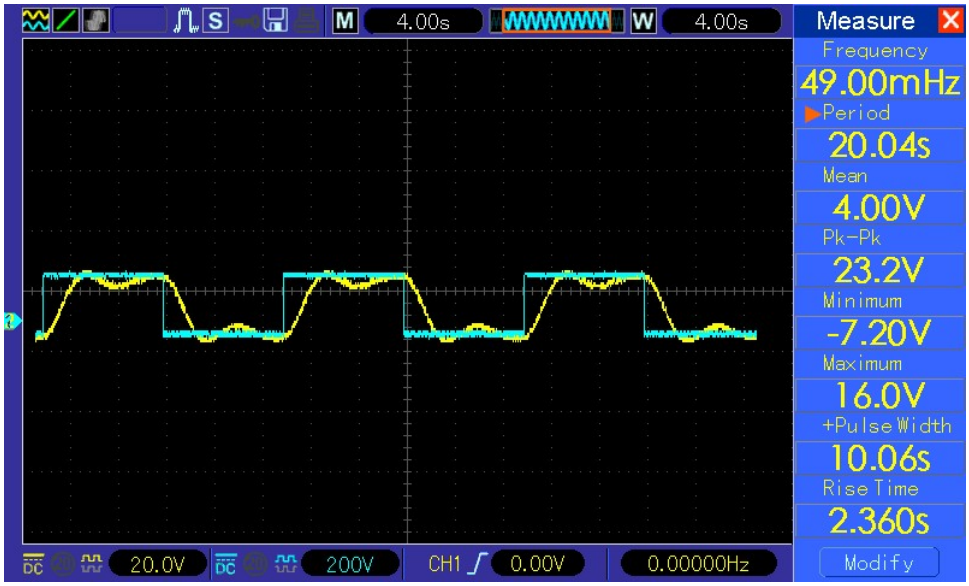


Figure 5.3: Closed loop suspension system response on dspace

The optimal gain matrix is determined through a process called "pole placement",which assigns specific locations for closed-loop poles to maintain system stability. Characteristics of the system's time response, such as rise time, settling time, and transient oscillations, are directly affected by the locations of the closed-loop poles.

Statespace response:

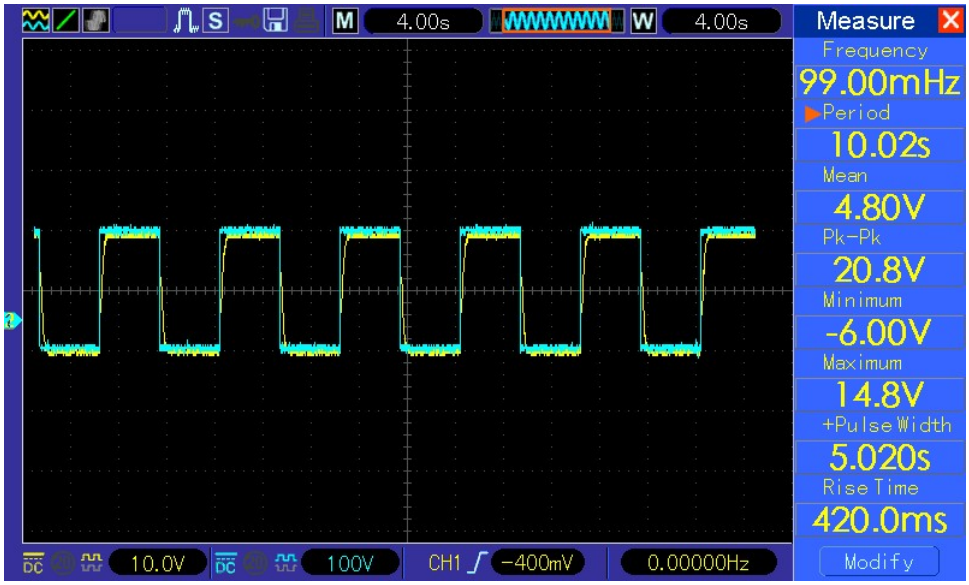


Figure 5.4: Statespace response of suspension system on dspace

5.2 Cruise Control System

Cruise Control System Open Loop Response:

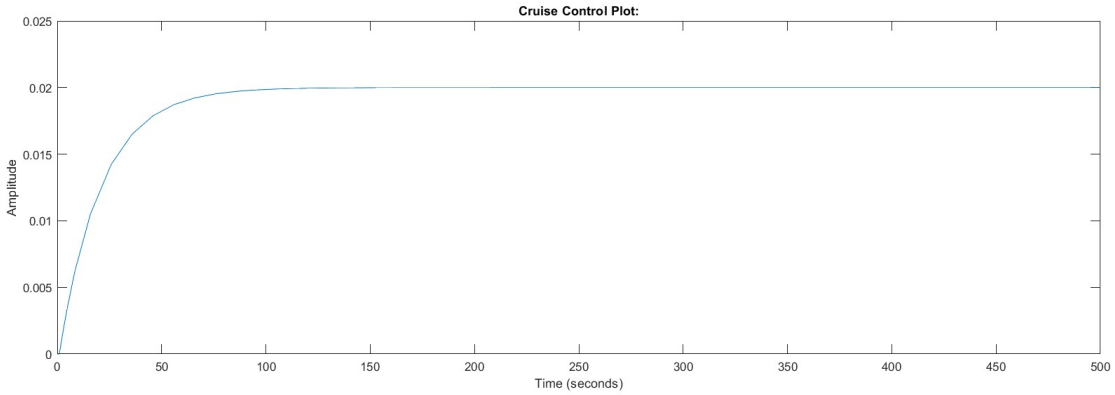


Figure 5.5: Open loop cruise control system response

Refer 4.7 for open loop block diagram. From the graph, we see that the open-loop system exhibits no overshoot or oscillations (which is characteristic of first-order systems) however the rise time is too slow. The open-loop system is stable and does not oscillate since the pole is real and negative. The cruise control transfer function equation is given as 1/(m*s+b) where m = 1000 and b = 50. From this equation we see the pole is -b/m. Furthermore, the speed of response is determined by the magnitude of this pole, b/m: the larger the magnitude, the quicker the system approaches the steady-state value. Since we're typically not able to change the system parameters thus to change the dynamic response of the system, we must instead design controllers which alter the poles and zeros of the closed-loop system to meet the desired performance specifications.

Therefore we need to design a feedback controller which speeds up the response significantly without negatively affecting the other dynamic performance metrics.

Cruise Control System Close Loop Response:

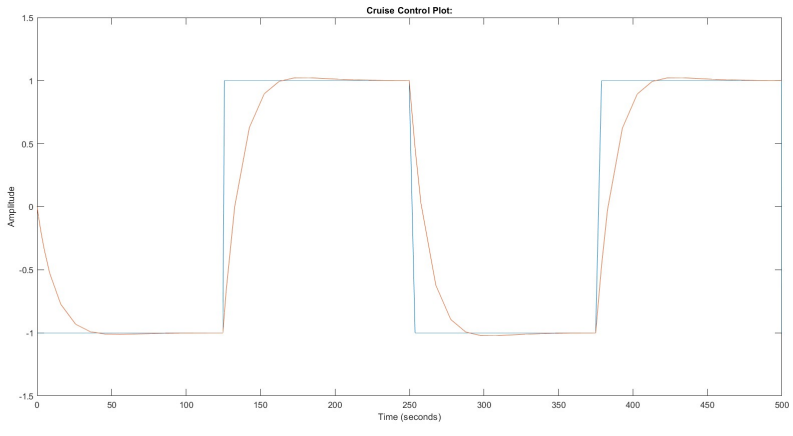


Figure 5.6: Closed loop cruise control system response

The steady-state error is now essentially zero, and the rise time has been reduced as seen from the graph. We have built a plant with PI controller (Refer 4.8 closed loop block diagram), From the closed loop response, we can see that the addition of an integral controller to the system eliminates the steady-state error and the rise time is also faster now. In the above graph, blue is input and red is output response.

Further, the real-time implementation of the closed loop system is done on dspace.

After successfully testing and applying the open loop and closed loop features of the individual transfer functions on Dspace and analyzing their frequency domain responses, the same process was repeated for the state-space model of the transfer function in the time domain. In order to examine the characteristics of the transfer function in the state-space, the strategy of pole placement was employed to ensure stability in the time domain.

Closed loop dspace response:

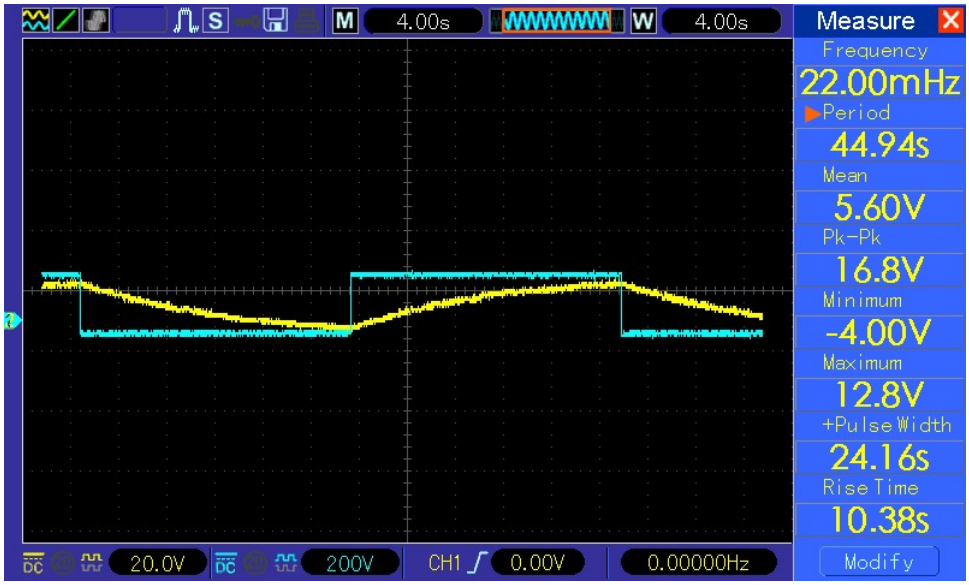


Figure 5.7: Closed loop cruise control system response on dspace

The optimal gain matrix is determined through a process called “pole placement”,which assigns specific locations for closed-loop poles to maintain system stability. Characteristics of the system’s time response, such as rise time, settling time, and transient oscillations, are directly affected by the locations of the closed-loop poles.

Statespace Response:

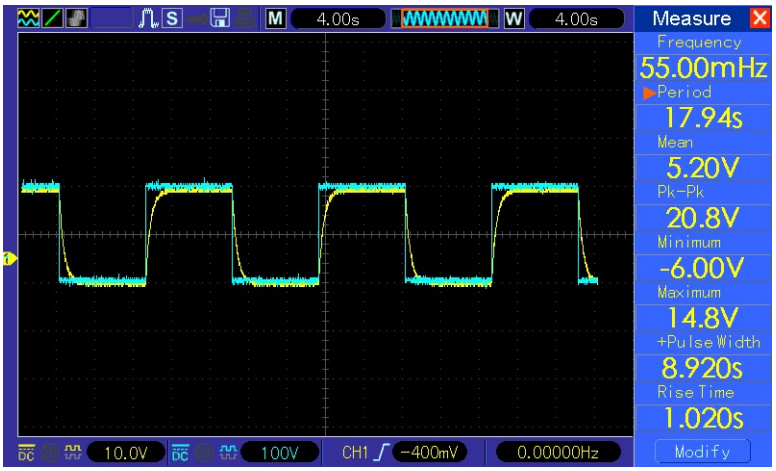


Figure 5.8: Statespace response of cruise control system on dspace

5.3 Antilock Braking System

Antilock Braking System Open Loop Response:

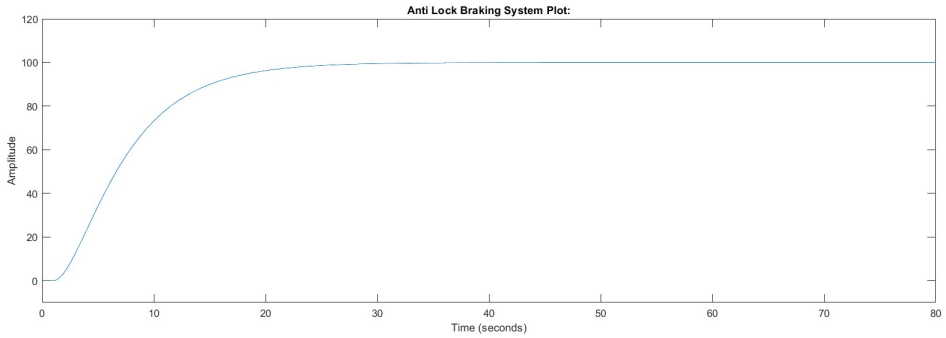


Figure 5.9: Open loop ABS response

Refer 4.13 for open loop block diagram. The system is stable because both poles have negative real parts. From the graph, we see that the open-loop system exhibits no overshoot or oscillations, however the rise time is slow. To reduce the response time and get optimal graph we design a controller plant.

Antilock Braking System Closed Loop Response:

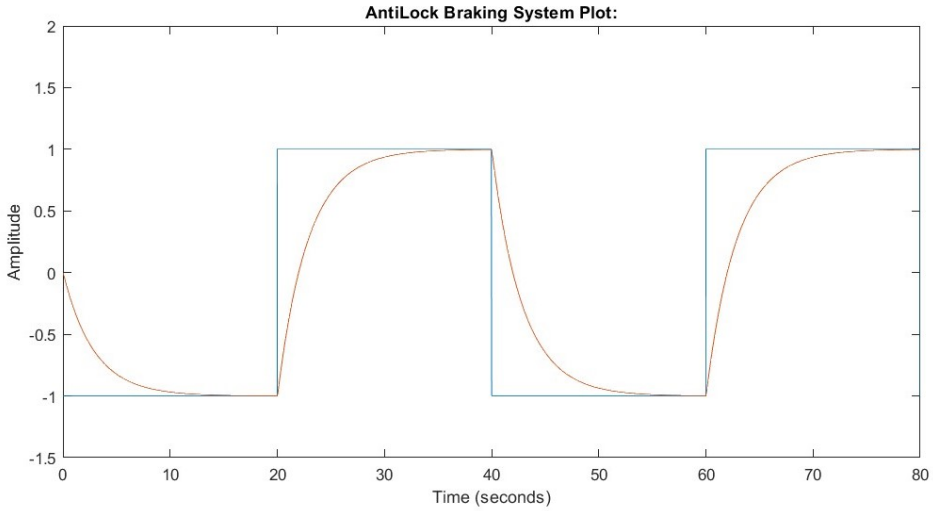


Figure 5.10: Closed loop ABS response

We have built a plant with PID controller (Refer 4.14 closed loop block diagram), the controller's parameter values were adjusted to optimize the output response in the frequency domain, ensuring both robustness and a faster settling time. The figures below depict the output graph, which appears to be relatively fast. In the above graph, blue is input and red is output response.

Further, the real-time implementation of the closed loop system is done on dspace.

After successfully testing and applying the open loop and closed loop features of the individual transfer functions on Dspace and analyzing their frequency domain responses, the same process was repeated for the state-space model of the transfer function in the time domain. In order to examine the characteristics of the transfer function in the state-space, the strategy of pole placement was employed to ensure stability in the time domain.

Closed Loop Dspace Response :

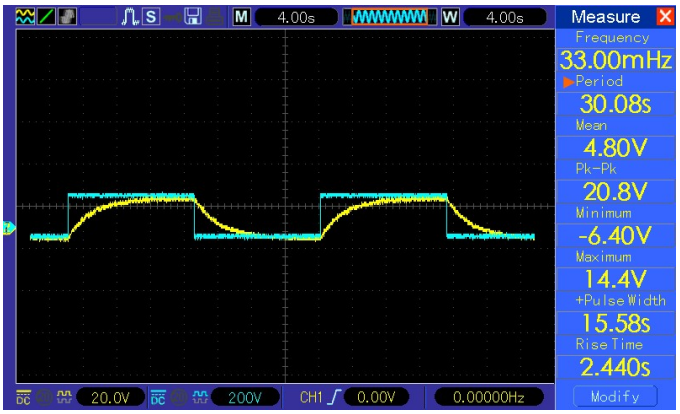


Figure 5.11: Closed loop ABS response on dspace

Statespace Response:

The optimal gain matrix is determined through a process called"pole placement", which assigns specific locations for closed-loop poles to maintain system stability. Characteristics of the system's time response, such as rise time, settling time, and transient oscillations, are directly affected by the locations of the closed-loop poles.

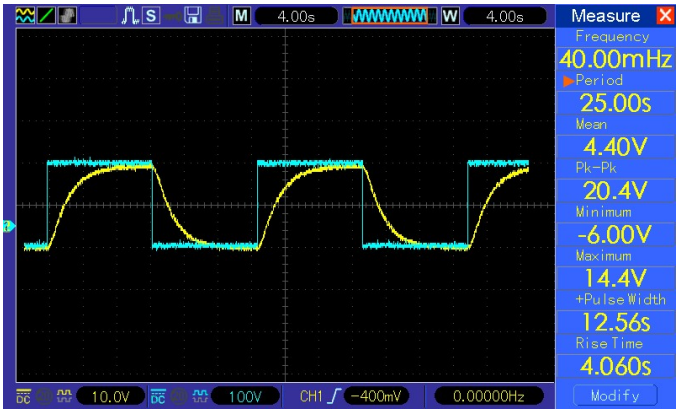


Figure 5.12: Statespace response of ABS on dspace

5.4 Motor Speed System

Motor Speed System Open Loop Response:

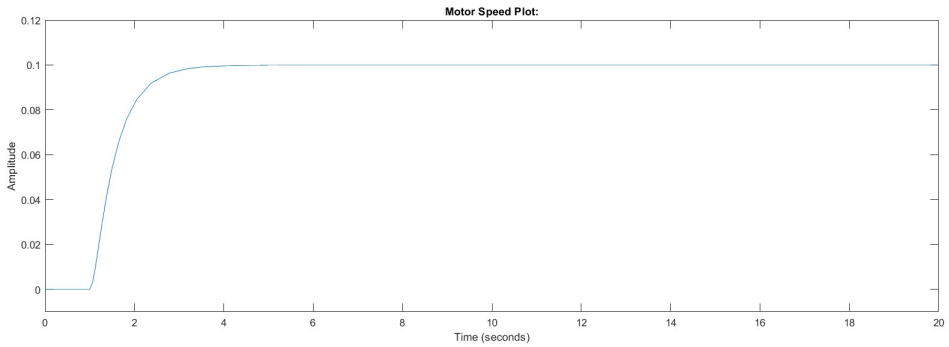


Figure 5.13: Open loop motorspeed system response

From the equation 4.4, you can see that the open-loop transfer function of motor speed has two negative real poles, one at s = -2 and one at s = -10. (Refer 4.19 for the open loop block diagram.) Since both poles are real, there is no oscillation in the step response (or overshoot) as we have already seen. Furthermore, since one pole is 5 times more negative than the other, the slower of the two poles will dominate the dynamics. That is, the pole at s = -2 primarily determines the speed of response of the system and the system behaves similarly to a first-order system.

Motor Speed System Close Loop Response:

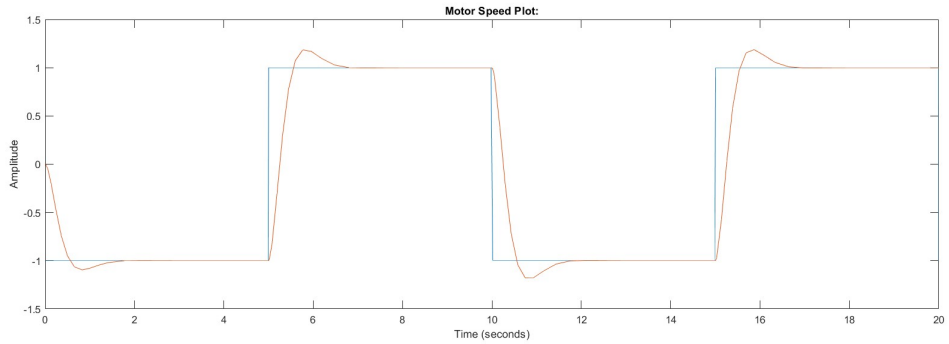


Figure 5.14: Closed loop motorspeed system response

We have built a plant with a PI controller (Refer 4.20 closed-loop block diagram), From the closed-loop response, we can see that adding an integral term will eliminate the steady-state error to a step reference. In the above graph, blue is input and red is output response.

Further, the real-time implementation of the closed loop system is done on dspace.

After successfully testing and applying the open loop and closed loop features of the individual transfer functions on Dspace and analyzing their frequency domain responses, the same process was repeated for the state-space model of the transfer function in the time domain. In order to examine the characteristics of the transfer function in the state-space, the strategy of pole placement was employed to ensure stability in the time domain.

Closed loop dspace Response:

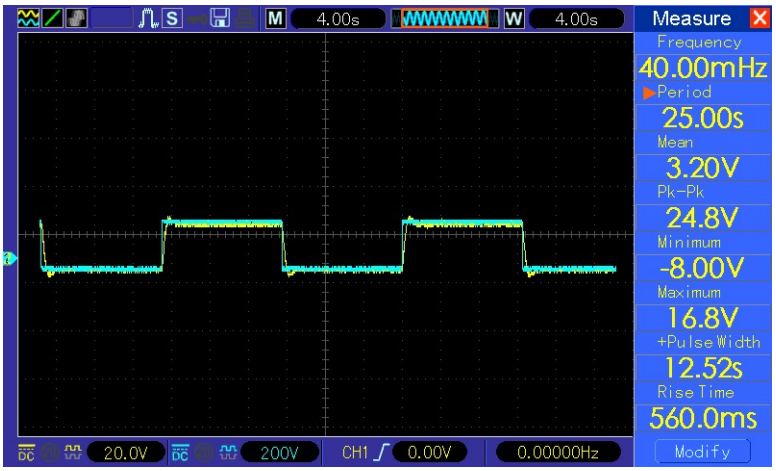


Figure 5.15: Closed loop motor speed system response on dspace

State Space Response:

The optimal gain matrix is determined through a process called "pole placement", which assigns specific locations for closed-loop poles to maintain system stability. Characteristics of the system's time response, such as rise time, settling time, and transient oscillations, are directly affected by the locations of the closed-loop poles.

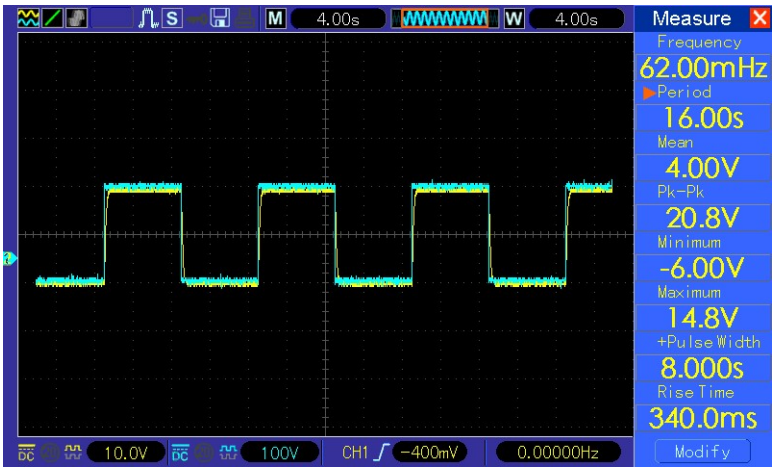


Figure 5.16: Statespace response of motor speed system on dspace

5.5 Throttle System

Refer figure 4.18 for open loop block diagram.

Throttle System Open Loop Response:

Refer 4.25 open loop block diagram. From the equation 4.21 we can calculate the poles of the transfer function. The poles are -0.5000 + 0.9220i , -0.5000 - 0.9220i The system is stable since both poles have negative real parts. For stability, it is important that all poles have negative real parts, which is the case here. The system has complex conjugate poles, indicating an underdamped response. The presence of complex conjugate poles suggests that the system will exhibit oscillatory behavior in its response. This can be overcome by designing a controller plant.

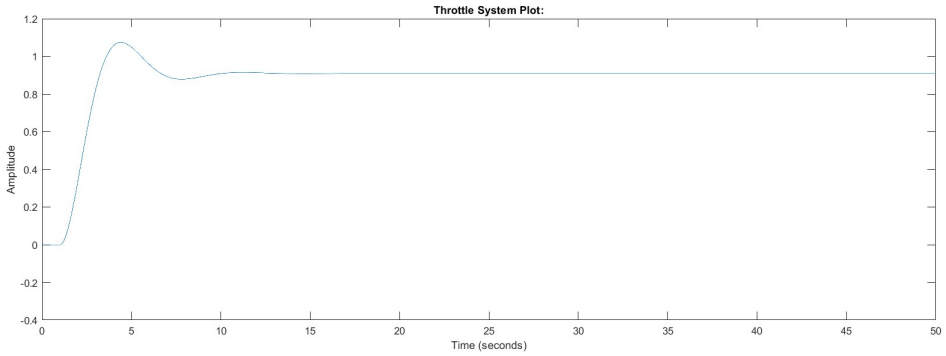


Figure 5.17: Open loop throttle system response

Throttle System Close Loop Response:

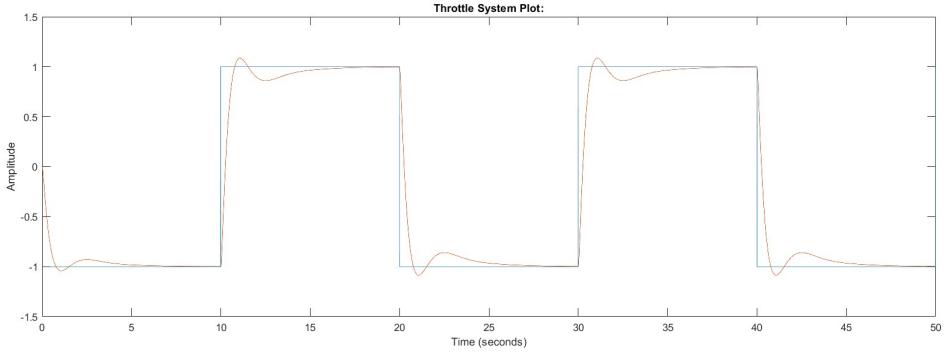


Figure 5.18: Closed loop throttle system response

We have built a plant with PID controller (Refer 4.26 closed loop block diagram), From the closed loop response, we can see that the amplitude of oscillation is reduced with less settling time. In the above graph, blue is input and red is output response.

Further, the real-time implementation of the closed loop system is done on dspace.

After successfully testing and applying the open loop and closed loop features of the individual transfer functions on Dspace and analyzing their frequency domain responses, the same process was repeated for the state-space model of the transfer function in the time domain. In order to examine the characteristics of the transfer function in the state-space, the strategy of pole placement was employed to ensure stability in the time domain.

Closed loop dspace Response:

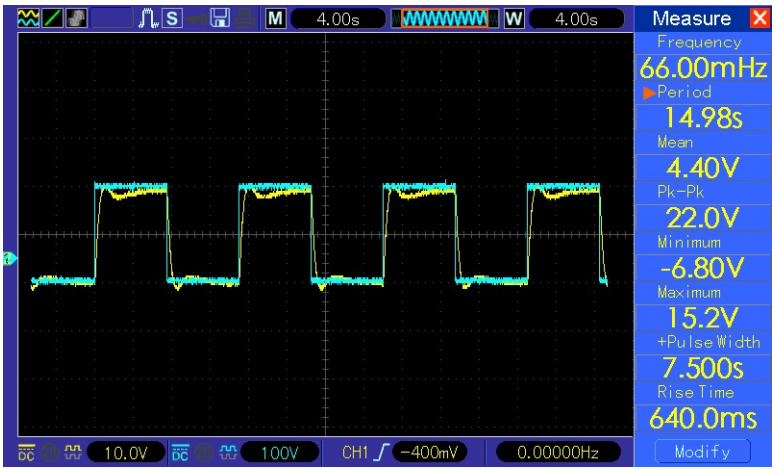


Figure 5.19: Closed loop throttle system response on dspace

State Space Response:

The optimal gain matrix is determined through a process called "pole placement", which assigns specific locations for closed-loop poles to maintain system stability. Characteristics of the system's time response, such as rise time, settling time, and transient oscillations, are directly affected by the locations of the closed-loop poles.

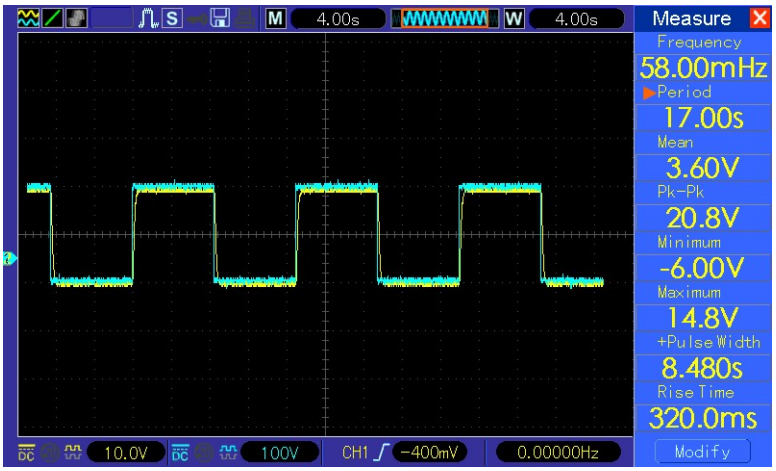


Figure 5.20: Statespace response of throttle system on dspace

Chapter 6

Conclusion and Future Implementation

6.1 Conclusion

In this work, our team has successfully developed a real-time simulation of vehicle dynamics and control systems. We have implemented transfer function models to effectively gain insight into the behavior of each system with respect to various variables. Through simulations conducted on the Microlabbox platform using DSpace as an application for development and testing, we have observed the effectiveness of these advanced control systems in improving both safety and performance. These results prove that utilizing such systems can help improve the overall capabilities of modern vehicles and provide automotive industry professionals with valuable insights into the design and implementation of closed-loop control systems.

Additionally, this project demonstrates how modern tools such as Microlabbox can be used for effective prototyping and testing purposes. All this data corroborates our initial objectives, showing that a real-time simulation is useful in understanding vehicle dynamics and control system behavior as well as producing superior results which can be applied within the automotive industry setting.

Overall, this project showcases the potential impact of advanced control systems on enhancing vehicle performance and safety. It serves as a foundation for further research and development in the field of vehicle dynamics and control systems, benefiting the automotive industry in its pursuit of improved driving experiences and safety measures.

6.2 Future Implementation

- Advanced Control Algorithms like model predictive control (MPC), adaptive control, or fuzzy logic control can the enhance the performance of control system.
- To enhance the implementation process, we can consider incorporating more extensive real-time testing by utilizing advanced Hardware-in-the-Loop (HIL) systems. This may entail integrating the control systems with more sophisticated HIL platforms that effectively replicate the vehicle's dynamics and environment, enabling thorough validation and evaluation of its performance.
- Controller and Plant are Implemented on different hardware platforms such as DSP processor/Microcontroller/FPGA and D-space

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Acknowledgments

My first and sincere appreciation goes to my H.O.D. and Project guide Prof. Vishwesh A. Vyawahare and Project Co-Ordinator Dr. Prassidh Trivedi for all I have learned from them and for their continuous help and support in all stages of this dissertation. I would also like to thank them for being an open person to ideas, and for encouraging me and my team to shape our interest and ideas.

I extend my gratitude towards my teachers, who equally motivated and directed me towards the path of knowledge thereby helping me to gain theoretical understandings of different concepts I studied till date.

I am thankful to our Principal Dr. M. D. Patil for his support and encouragement. My thanks and appreciations also go to my teammates in developing the project and people who have willingly helped me out with their abilities. I cannot end without thanking my lovely family for their encouragement.

I am grateful towards all the people who directly and indirectly helped me for this project

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Date Signature